AN ILLUSTRATED CLINICAL GUIDE

BIOCERAMIC IN DENTISTRY

Expanded & Illustrated Edition



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PRINCE ABDULRAHMAN ADVANCED DENTAL INSTITUTE

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This edition preserves the text and scholarship of the original work by Dr. Saad Alsaif and Dr. Khalid Al Otaibi (Prince Abdulrahman Advanced Dental Institute, Riyadh, 2026) and augments it with explanatory expansion, summary call-outs, comparative data charts and original schematic illustrations.

The illustrations are schematic diagrams created to clarify concepts and material behaviour. They are not clinical photographs, radiographs or histology, and are not drawn to scale. Numerical values in the charts are representative figures drawn from the studies cited in the text and are intended for orientation.

How to read this edition

- Teal call-outs summarise the key take-aways of a section.
- Gold call-outs flag practical or clinical points.
- Figures are numbered per chapter; tables retain the original numbering.

Four further chapters broaden the scope beyond endodontics. Chapter Five, ‘Bioceramics across Dentistry’, and Chapter Six, on properties, testing and handling, are expository background compiled for this edition; they summarise well-established dental-materials science and are presented without new author–year citations so that no source is misattributed. Chapter Seven, on dental adhesives, and Chapter Eight, on the periodontal response to materials, additionally draw on supplied sources and carry author–year citations, listed under ‘Additional references’. The main References section reproduces the full bibliography of the original work, so that every citation in the endodontic chapters can be traced to its source.

Clinical decisions should always be based on current primary literature and individual patient assessment.

Preface

Few groups of materials have changed a dental discipline as quickly as bioceramics changed endodontics. In the span of a single generation, mineral trioxide aggregate turned procedures that were once uncertain — sealing a perforation, capping an exposed pulp, treating a tooth with an open apex — into predictable, biologically driven treatments. The materials that followed have widened that success and carried the same chemistry into restorative dentistry, implantology and bone regeneration.

The original monograph on which this edition is based set out the science of these materials for the endodontic clinician with clarity and rigour. This expanded edition keeps that foundation intact and adds three things: illustration, so that mechanisms and clinical situations can be seen as well as read; breadth, through two new chapters that place endodontic bioceramics in the wider context of dentistry and of materials science; and reference, through the full bibliography of the original work and a set of quick-reference appendices.

The aim throughout has been to make the subject easier to learn without diluting its accuracy. The illustrations are schematic, the comparative figures are drawn from the cited literature, and the new background chapters are written as established materials science rather than as novel claims. Where the evidence is still incomplete — as it often is for long-term clinical outcomes — the text says so plainly.

It is hoped that students, residents and practising clinicians alike will find here both a readable introduction and a practical point of reference.

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CHAPTER ONE

Introduction

Bioceramics in medicine, dentistry and endodontics

Bioceramics in medicine

Over the past several decades the materials available to medicine have advanced dramatically, and ceramic materials in particular have been engineered to repair and reconstruct the diseased or damaged parts of the musculo-skeletal system. Ceramics used for this purpose are termed bioceramics (Hench, 1991). Since the 1960s, alongside advances in technology and knowledge, this family has grown to include alumina, zirconia, coatings and composites, hydroxyapatite, bioactive glass, glass-ceramics, resorbable calcium phosphates, radiotherapy glasses and calcium silicate (Best et al., 2008; Dubok, 2000; Hench, 1991).

Hench (1991) and Best et al. (2008) classified bioceramics as bioinert, bioactive and biodegradable (soluble or resorbable). In practice no implant material is completely inert: every material provokes at least a minimal reaction in living tissue (Dubok, 2000). A biomaterial, more broadly, is any synthetic or natural material used to perform or substitute a body function or tissue (Bandyopadhyay and Bose, 2013).

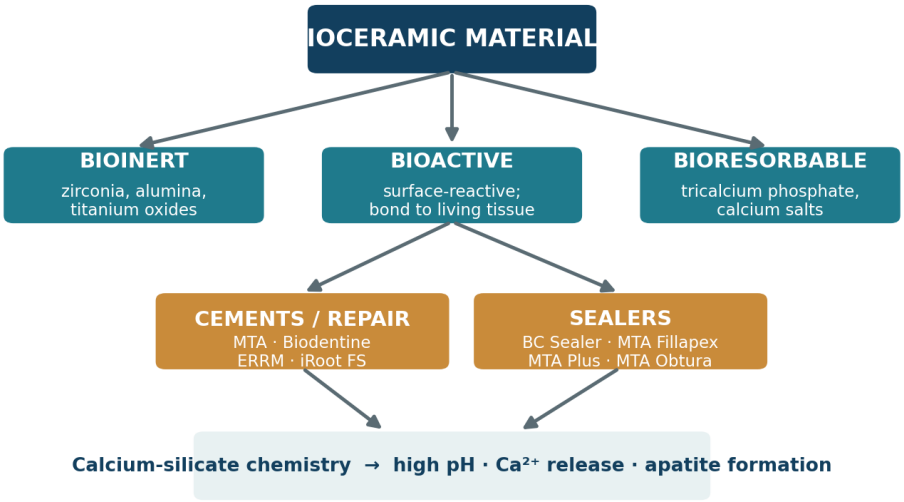


Figure 1.1 Bioceramics are grouped by how they interact with tissue; endodontic materials are predominantly bioactive calcium-silicate cements and sealers.

Biomaterials can be described as biotoxic, bioinert, bioactive or bioresorbable. Biotoxic materials — certain cadmium- or vanadium-containing alloys, carbon steel and methyl methacrylate — provoke pathological change or immune rejection. Bioinert materials such as aluminium and zirconium oxides, tantalum and titanium coexist with tissue, separated only by a thin fibrous layer. Bioactive materials — high-density hydroxyapatite, tricalcium phosphate and certain bioglasses — form direct biochemical bonds with tissue. Bioresorbable materials are gradually resorbed and replaced by healthy tissue as the wound heals (Dubok, 2000).

At a glance

- Bioceramics are the only synthetic inorganic materials that can be both bioactive and bioresorbable.
- They provoke less immune response and offer a wider range of mechanical and biochemical behaviour than most implants.
- Their main limitation is low crack resistance — they are brittle and hard to adjust intra-operatively.

Because bioceramics are among the few inorganic materials that demonstrate both bioactive and bioresorbable behaviour, and because their biocompatibility exceeds that of most implanted materials, they have been used extensively across surgical medicine (Dubok, 2000; Hench, 1991). Their principal weakness relative to metals is low crack resistance: they are brittle and difficult to manipulate. Combining bioceramic particles with metallic implants captures the advantages of both — carbon coatings, for example, are thromboresistant and are used on prosthetic heart valves and orthopaedic and maxillofacial prostheses (Geesink and Manley, 1993; Hench, 1991).

Bioinert ceramics retain their characteristics indefinitely within the body, which makes them valuable for immunocompromised and oncological patients. Aluminium oxide (corundum) was the first widely used bioceramic thanks to its high wear resistance, strength and corrosion resistance; it is used in load-bearing hip prostheses, dental implants, knee prostheses, alveolar-ridge and maxillofacial reconstruction, bone screws, and even replacements for the cornea and the ossicles of the

middle ear (Hench, 1991; Hulbert et al., 1987). Despite its compatibility, alumina has low crack resistance; zirconium dioxide offers similar biocompatibility with greater bend strength and crack resistance, especially when stabilised with yttrium or cerium oxide (Dubok, 2000).

Bioactive and bioresorbable ceramics are used to aid healing of bone defects after removal of cysts and tumours: bioactive ceramics integrate with bone and hold their strength during healing, while bioresorbable ceramics are gradually replaced by new bone. Between them they serve maxillofacial reconstruction, cranial repair, bone grafting, joint reconstruction, hearing endoprostheses and cosmetic eye prostheses; bioresorbable ceramics can also support restoration of tendons, ligaments, nerve fibres and blood vessels (Black, 1988; Hulbert et al., 1987). The surface biochemical bonding that defines bioactivity was first detected in the late 1960s (Hench et al., 1971).

Bioceramics in dentistry

Bioceramics are not new to dentistry. Hulbert et al. (1987) reviewed calcium-phosphate bioceramics used more than two decades earlier for periodontal treatment, alveolar-ridge augmentation, maxillofacial surgery and dental implants. Hench (1991) noted the use of alumina, hydroxyapatite and bioactive glasses for dental implants and maxillofacial reconstruction, and of hydroxyapatite, tricalcium phosphate, calcium-phosphate salts and bioactive glasses for periodontal-pocket obliteration. Their excellent physicochemical and biological properties made these materials natural candidates for endodontics.

Bioceramics in endodontics

Bioceramics entered endodontics in the 1990s in the form of mineral trioxide aggregate (MTA), first used as a retrograde (root-end) filling and later for root repair and as a coating for gutta-percha cones (Koch and Brave, 2012). Advances in nanotechnology have since widened their use to root-canal sealing. The reasons for choosing them are compelling: bioceramics are highly biocompatible, non-toxic, chemically stable in a biological environment and do not shrink on setting. An overfill during obturation or root repair does not provoke an inflammatory reaction, and

the materials form hydroxyapatite that strengthens the bond to dentine (Koch and Brave, 2012).

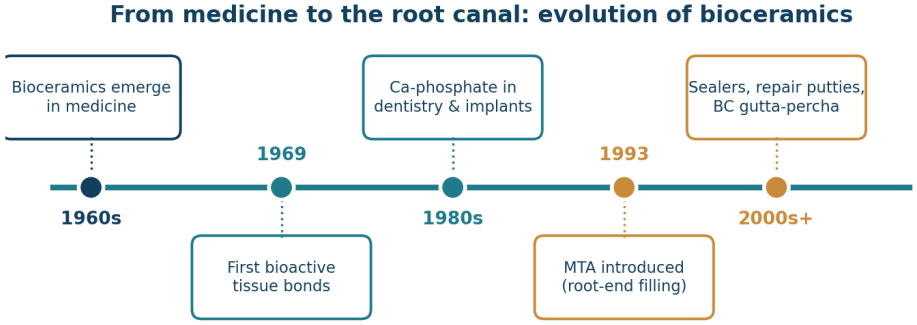


Figure 1.2 Bioceramics migrated from general medicine to dentistry and, from 1993, into routine endodontic practice.

Bioceramics are notably forgiving in technique because blood and moisture do not contaminate them; indeed they require moisture to set (Hench, 1991; Jefferies, 2014). They offer excellent sealing because they remain dimensionally stable and expand marginally on setting (Gandolfi et al., 2009; Jefferies, 2014). On setting, the hydration reaction liberates calcium hydroxide, which dissociates into calcium and hydroxyl ions and raises the pH above 12 (Jefferies, 2014; Richardson, 2008). While unset they are antibacterial; once set they become biocompatible and bioactive, forming hydroxyapatite in tissue fluids as released calcium hydroxide reacts with phosphates (Richardson, 2008). This apatite formation underlies their tissue-inductive behaviour.

Why bioceramics changed endodontics

- Not moisture-sensitive — they set in the wet field of the root canal.
- High pH on setting → antibacterial while unset; bioactive once set.
- Dimensionally stable with marginal expansion → excellent seal, no shrinkage.
- Form hydroxyapatite and bond to dentine → tissue-inductive healing.

The technology has changed how endodontists plan treatment, surgically and non-surgically, improving the prognosis of teeth that would once have been lost. Bioceramics are now a first choice for pulpotomy, pulp

capping, perforation repair, root-end filling, obturation of immature (open-apex) teeth and canal sealing in mature teeth. Commercial products include cements — Biodentine, the various MTA brands, EndoSequence Root Repair Material (ERRM) putty and paste, and iRoot FS — and sealers such as EndoSequence BC Sealer, iRoot SP, MTA Fillapex, MTA Plus and bioceramic gutta-percha. The chapters that follow examine these materials in turn.

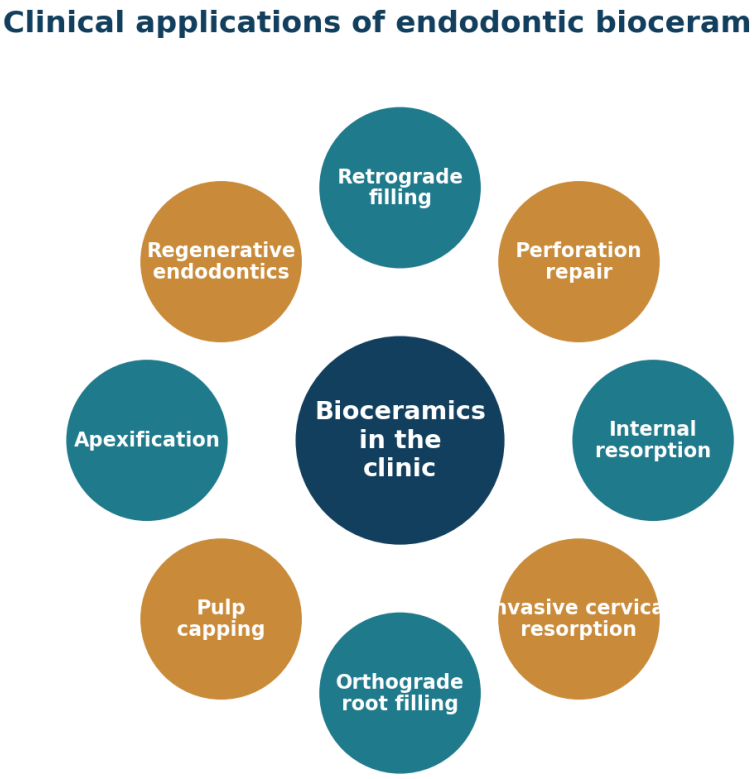


Figure 1.3 The clinical territory covered by bioceramics now spans surgical and non-surgical endodontics — each application is discussed in Chapter Four.

CHAPTER TWO

Bioceramic Cements

MTA · MTA Angelus · Biodentine · ERRM & iRoot

Bioceramic cements are the bulk-placement members of the family: they are packed into a cavity, perforation or canal terminus to form a solid, sealing mass. The prototype is MTA, and the many products that followed refine its handling, setting time and aesthetics while retaining its calcium-silicate chemistry.

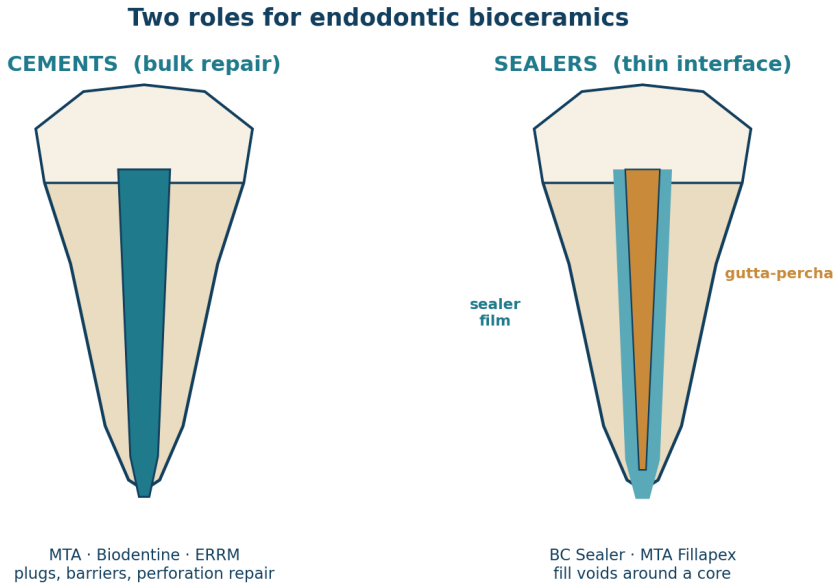


Figure 3.1 Cements are placed in bulk to form barriers and plugs; sealers (Chapter Three) form a thin film around a core material.

An optimal endodontic material

Before comparing individual products it helps to state the target. Wang (2015) summarised the physicochemical and biological characteristics of an ideal endodontic material (Table 2):

Physicochemical characteristics	Biological properties
Short setting time	Biocompatible
High mechanical strength	Stimulates biomineralisation
High alkalinity (pH) & Ca^{2+} release	Induces pulp-cell differentiation
Marked radiopacity	Antibacterial activity
Moderate flow, low porosity & solubility	

Table 2. Characteristics of an optimal material used in endodontics (Wang, 2015).

2.1 Mineral trioxide aggregate (MTA)

MTA powder is essentially a purified Portland cement: tricalcium silicate, dicalcium silicate, tricalcium aluminate, calcium sulfate dihydrate (gypsum), calcium aluminoferrite and bismuth oxide. It is mixed with sterile water at roughly a 3:1 powder-to-water ratio (Torabinejad et al., 1993). Two reactions drive setting. In the first, the silicate phases hydrate to form calcium silicate hydrate (C–S–H) and calcium hydroxide; the calcium hydroxide raises pH, confers antibacterial activity and seeds the deposition of mineralised apatite. In the second, tricalcium aluminate reacts with water in the presence of calcium sulfate to form ettringite, and — once the sulfate phases are consumed — monosulfate (Camilleri et al., 2013).

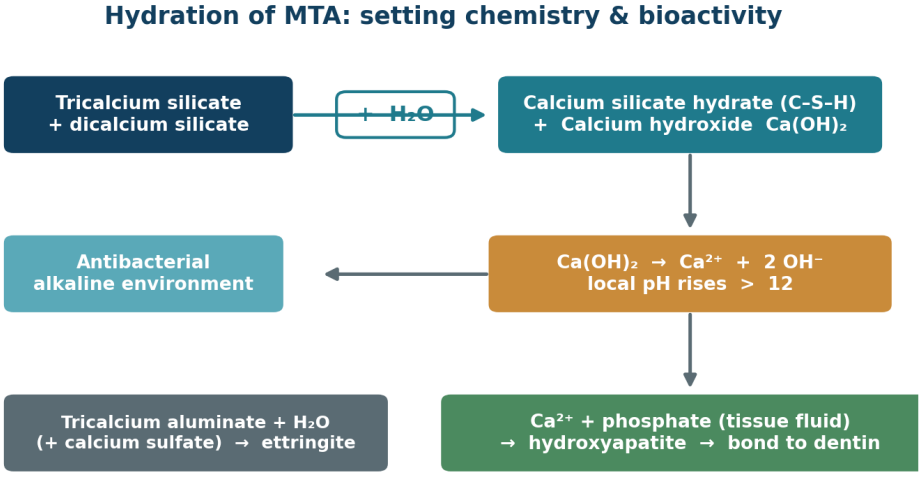


Figure 2.1 The silicate (first) reaction generates the high pH and apatite-forming behaviour that make MTA bioactive; the aluminate (second) reaction governs early set.

Key point — moisture is a friend, not an enemy

- Most dental materials degrade in a wet field. MTA is the opposite: it requires moisture to set, which is why it was introduced to overcome contamination problems (Shah et al., 1996).

MTA is sold in grey (GMTA) and white (WMTA) forms; WMTA was developed to reduce the dentine discolouration seen with the original grey material (Kratchman, 2004). Its chief drawback is a long setting time — a mean of about 165 minutes for conventional MTA (Gandolfi et al., 2009; Torabinejad et al., 1995a) — although WMTA sets faster than GMTA (Islam et al., 2006). A long set makes it hard to maintain a consistent mix, whereas a short set favours a tight seal between canal and periodontium (Santos et al., 2008).

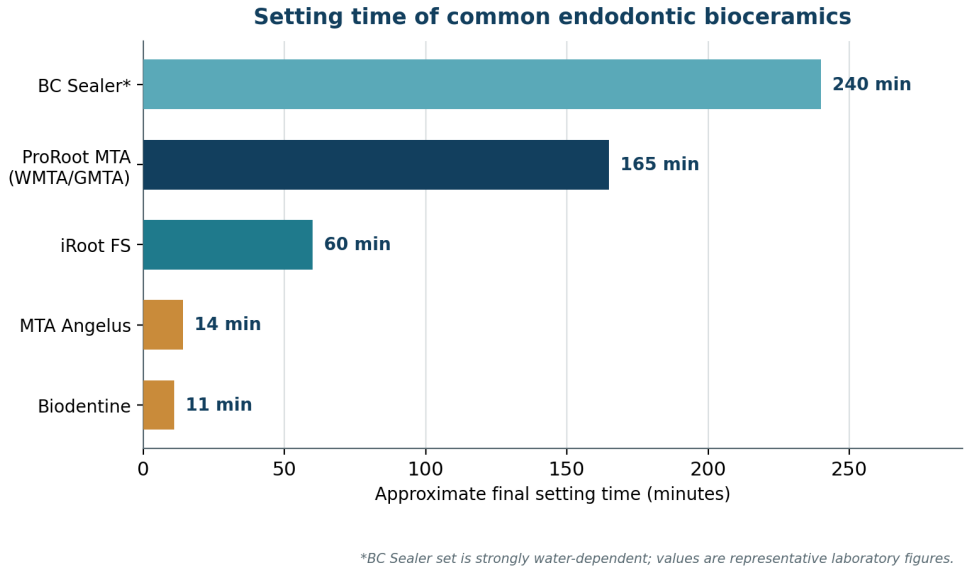


Figure 2.2 Conventional MTA sets slowly; later products such as MTA Angelus and Biodentine were engineered for a much shorter set.

Any material that permanently seals a canal should be alkaline and should release calcium ions to promote mineralised-apatite deposition — both in-vitro and in-vivo work links pulp wound healing to an alkaline pH and calcium-ion availability (Holland et al., 2001; Okabe et al., 2006). The pH of MTA rises from about 10.2 immediately after mixing to 12.5 by three hours (Torabinejad et al., 1995a); GMTA shows a lower value at 60 minutes than WMTA (Chng et al., 2005). The high pH reflects continuous liberation of calcium ions and formation of calcium hydroxide (Fridland and Rosado, 2005; Han and Okiji, 2013).

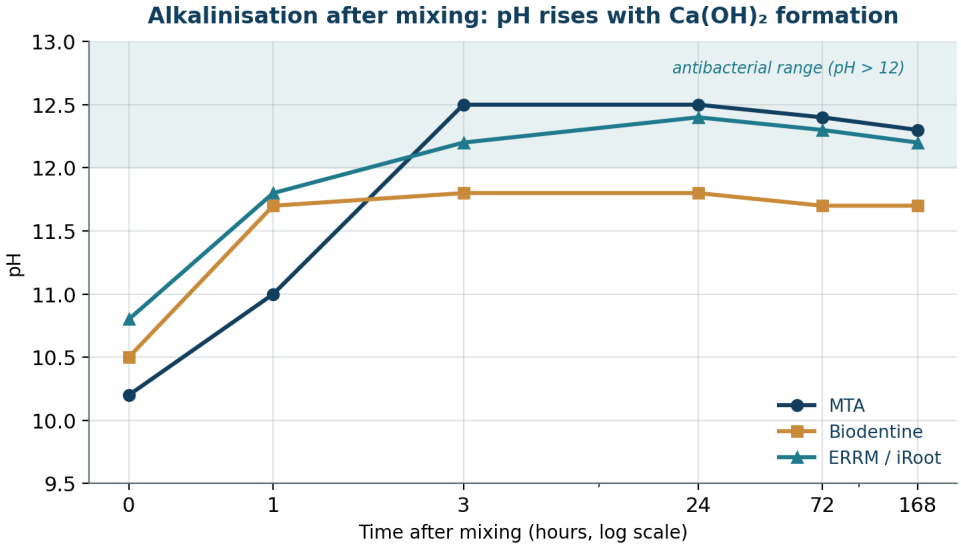


Figure 2.3 Alkalinisation follows Ca(OH)₂ formation and is shared, to differing degrees, by the major cements.

Radiopacity lets the clinician confirm the quality of a filling radiographically (Torabinejad et al., 1995a). Bismuth oxide is added to calcium-silicate cement to raise radiopacity (Saghiri et al., 2015); under ISO 6876/2001 a material must be at least as radiopaque as 3 mm of aluminium. MTA reaches about 7.17 mm (Torabinejad et al., 1995a), with WMTA slightly higher than GMTA despite equal bismuth content (Parirokh and Torabinejad, 2010a).

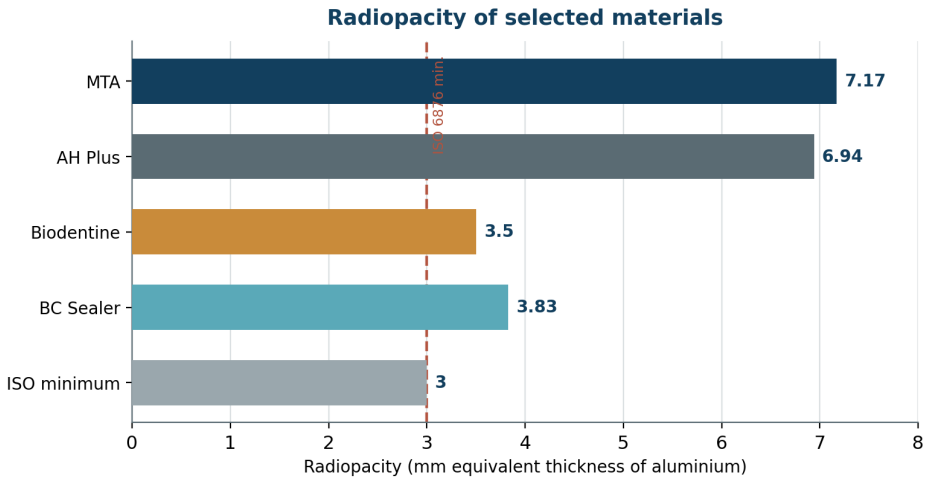


Figure 2.4 All the calcium-silicate materials clear the ISO minimum; MTA and AH Plus are the most radiopaque of those shown.

Solubility should be low for root-end and repair materials; MTA showed little or no solubility in most studies, though one long-term study reported an increase (Fridland and Rosado, 2005; Poggio et al., 2007; Shie et al., 2009). Porosity should also be low, since mechanical strength rises as porosity falls (Holt et al., 2007); an acidic environment and air bubbles introduced during mixing both raise porosity (Namazikhah et al., 2008), while chelating agents, low humidity, high condensation forces and low pH can reduce microhardness (Nekoofar et al., 2007).

Compressive strength depends on the type of MTA and is higher for conventional MTA than for MTA Angelus (Basturk et al., 2013); phosphoric-acid etching lowers it (Kayahan et al., 2009). Amalgam, IRM and Super-EBA initially exceed MTA in compressive strength, but the difference disappears by three weeks (Torabinejad et al., 1995a). Placing a wet cotton pellet over MTA for a day can raise its flexural strength by exposing it to moisture on two sides (Walker et al., 2006). Push-out bond strength is pH-sensitive: it is highest near pH 8.4 and falls in more acidic conditions (Saghiri et al., 2010; Shokouhinejad et al., 2010).

Biomineralisation arises from the slow release of calcium ions during setting, which up-regulates signalling molecules — transforming growth factor-β, macrophage colony-stimulating factor and interleukins — that recruit progenitor cells and drive their differentiation into

odontoblast-like cells and dentinogenesis (Koh et al., 1997; Okiji and Yoshiba, 2009). MTA can induce an osteogenic phenotype, up-regulating osteonectin, osteopontin and alkaline phosphatase (Bonson et al., 2004), and the resulting mineral tags within dentinal tubules improve push-out bond strength (Reyes-Carmona et al., 2010).

Biocompatibility is well documented: MTA outperformed amalgam, IRM, Super-EBA, gutta-percha and several glass ionomers in cytotoxicity and cell-attachment studies (Osorio et al., 1998; Souza et al., 2006; Yoshimine et al., 2007) and is comparable to the newer bioceramics — BC Sealer, ERRM and Biodentine (Ma et al., 2011; Zhang et al., 2010b; Zhou et al., 2013). Reports on its antibacterial effect conflict, but consensus holds that its high pH and calcium-hydroxide formation give it real antimicrobial activity, with differences between studies explained by the type and amount of MTA and the organisms tested (Parirokh and Torabinejad, 2010a).

2.2 MTA Angelus

Developed in Brazil, MTA Angelus is 80% Portland cement and 20% bismuth oxide, with the calcium sulfate removed to shorten the set (Duarte et al., 2003). Its setting time of about 14 minutes is far shorter than WMTA or GMTA (Chng et al., 2005; Santos et al., 2008). Its finer, more variable particle size (48 µm vs 65 µm for MTA) improves handling (Komabayashi and Spångberg, 2008). Reports on pH and calcium-ion release conflict — some find lower values than MTA, some higher, some no difference — probably because measurements were taken at different times after mixing (Cavenago et al., 2014; Duarte et al., 2003; Reyes-Carmona et al., 2009). Its radiopacity is lower than WMTA and GMTA (Chng et al., 2005).

MTA Angelus is biocompatible: neither grey nor white forms were genotoxic or cytotoxic to Chinese-hamster ovary or lymphoma cells at 1–1000 µg/mL (Ribeiro et al., 2006a, 2006b). It stimulated mineralisation in rat connective tissue (Gomes-Filho et al., 2009) and — with GMTA and Portland cement — was effective against the organisms tested, including E. coli (Tanomaru-Filho et al., 2007).

2.3 Biodentine

Biodentine combines tricalcium and dicalcium silicate with calcium carbonate and a water-based liquid containing calcium chloride as a setting accelerator. It sets in about 10–12 minutes and exceeds 3 mm-aluminium radiopacity (Grech et al., 2013b), which suits it to perforation repair. It is marketed as a dentine substitute for perforation repair and coronal restoration and can be used for pulp capping (File, 2010). Rheometry shows it has the shortest set among tricalcium-silicate cements (Setbon et al., 2014), and it produces only slight clinical discolouration — an advantage over MTA in aesthetic regions (Keskin et al., 2015).

Clinical note — irrigation and MTA strength

- Biodentine, amalgam and IRM keep their push-out strength after soaking in NaOCl, chlorhexidine (CHX) or saline, but MTA loses strength after CHX exposure (Guneser et al., 2013).
- Where MTA is planned, avoid CHX as the final irrigant to preserve bond strength.

Immersed in Hank's balanced solution, Biodentine held a pH of about 11.7 with little change over 28 days and released more calcium than MTA, BC Sealer, BioAggregate, IRM and TCS-Zr (Grech et al., 2013a; Han and Okiji, 2013). Its microhardness, compressive strength and push-out strength are high (Guneser et al., 2013). A small particle size (~5 µm) and homogeneous composition let it diffuse into dentinal tubules and form mineral-infiltration zones and tag-like structures, giving greater push-out strength than MTA and a cohesive rather than adhesive failure mode (Atmeh et al., 2012; Guneser et al., 2013).

Biocompatibility studies broadly favour Biodentine and MTA over glass ionomer, though some cultured cell lines showed lower viability with Biodentine than MTA at day seven (Pérard et al., 2013; Zhou et al., 2013) and both lowered proliferation of OD-21 pulp cells relative to control (Luo et al., 2014). Such discrepancies underline the need for well-designed, randomised long-term clinical trials; in-vitro and in-vivo evidence is generally favourable but long-term clinical data remain limited.

2.4 ERRM Putty, ERRM Paste and iRoot FS

EndoSequence Root Repair Material (ERRM) is supplied as a pre-mixed pliable putty (also branded iRoot BP Plus) and as a syringe-delivered paste for intracanal placement (Shi et al., 2016). Both share a composition of calcium silicate, tantalum oxide, zirconium oxide and monobasic calcium phosphate. Working time exceeds half an hour and setting takes roughly four hours, requiring moisture; the pH during setting can reach 12.4, contributing to antibacterial activity (Wang, 2015).

ERRM's compressive strength is similar to MTA (Walsh et al., 2014); an acidic environment makes both putty and paste more porous and less crystalline (Wang et al., 2015). iRoot FS (Fast Set) shares ERRM's composition but hardens within an hour at 37 °C in 100% humidity, with improved handling (Jiang et al., 2014). Cytotoxicity of ERRM is as low as MTA and MTA Angelus (Ciasca et al., 2012; Ma et al., 2011); iRoot BP Plus is non-toxic to human pulp cells with a pulp response indistinguishable from MTA (Shi et al., 2016), and iRoot FS showed the best cell-adhesion capacity of the materials compared (Jiang et al., 2014). In root-end microsurgery in dogs, ERRM Putty and MTA healed comparably (Chen et al., 2015), and both behaved similarly against planktonic E. faecalis (Lovato and Sedgley, 2011).

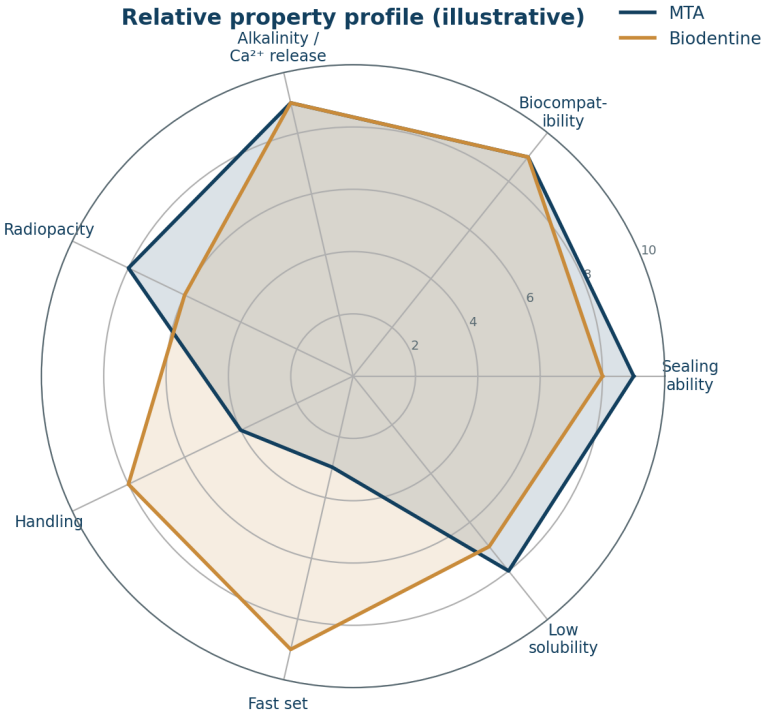


Figure 2.5 An illustrative property profile: Biodentine trades some radiopacity for faster set and easier handling, while MTA remains the reference for sealing and bioactivity.

CHAPTER THREE

Bioceramic Sealers

BC Sealer · MTA Fillapex · MTA Plus · MTA Obtura

Where cements are placed in bulk, sealers flow into the space between a core obturation material and the canal wall, filling voids and lateral anatomy. Bioceramic sealers bring the calcium-silicate advantages — high pH, calcium release, apatite formation and moisture tolerance — to routine canal obturation. They are grouped as pure bioceramic sealers and hybrid (resin-containing) sealers (Shen et al., 2015).

Pure bioceramic sealers	Hybrid bioceramic sealers
EndoSequence BC Sealer / iRoot SP	MTA Fillapex
TechBiosealer endo	MTA Plus
BioRoot RCS	MTA Obtura
Endo-CPM-Sealer	
ProRoot Endo Sealer	

Table 3. Bioceramic and hybrid sealers used in endodontics (Shen et al., 2015).

3.1 BC Sealer (iRoot SP)

EndoSequence BC Sealer, known as iRoot SP, is a ready-mixed sealer of tricalcium and dicalcium silicate, monobasic calcium phosphate, zirconium oxide, colloidal silica and calcium hydroxide (Hess et al., 2011). Zirconium oxide gives it a radiopacity of 3.83 mm aluminium — above the 3.00 mm minimum but below AH Plus at 6.94 mm (de Miranda Candeiro et al., 2012; Loushine et al., 2011). It is pre-mixed with water and free of liquid-thickening carriers; both absorbed water and water generated by the reaction between calcium phosphates and calcium hydroxide feed the calcium-silicate-hydrate phase, within which hydroxyapatite precipitates and reinforces the sealer–dentine bond (Loushine et al., 2011; Yang et al., 2002). Because of this it is difficult to remove completely from canals (Hess et al., 2011; Ma et al., 2012).

Clinical note — dry the canal, restore the crown

- BC Sealer's set and microhardness depend on water balance: too much water lengthens the set and softens the material (Loushine et al., 2011).
- Dry the canal before filling and provide a sound coronal restoration to avoid leakage and material deterioration.

BC Sealer has higher pH, calcium-ion release and flow than AH Plus (Loushine et al., 2011). Reports on bond strength vary — higher than AH Plus, Epiphany and MTA Fillapex in one study (DeLong et al., 2015); similar to AH Plus in others (Ersahan and Aydin, 2010; Sagsen et al., 2011); lower than AH Plus in another (Amin et al., 2012) — differences attributed to obturation technique and sealer brand. Solubility findings likewise differ (Borges et al., 2012; Ersahan and Aydin, 2010). Cytotoxicity depends on time and concentration: moderate when fresh, slight once set, and absent after long-term setting and high dilution (Mukhtar-Fayyad, 2011; Zhang et al., 2010b). iRoot SP induces biomineralisation and hard-tissue deposition (Güven et al., 2013; Zhang et al., 2010a) and is antibacterial against E. faecalis, with 18%, 35% and 45% of dentinal-tubule bacteria killed at 1, 7 and 30 days (Wang et al., 2014; Zhang et al., 2009).

3.2 MTA Fillapex

MTA Fillapex was designed to merge MTA's biological behaviour with the handling of a resin-based sealer (Silva et al., 2013). It offers acceptable working time and high radiopacity (though lower than AH Plus) and much higher flow than AH Plus (Silva et al., 2013; Xuereb et al., 2015). Its pH rose to about 10.06 after 15 hours in water, favouring apatite deposition and antibacterial activity, but it had the highest solubility of the sealers examined (Faria-Júnior et al., 2013) — a long-term disadvantage because solubility degrades the seal and permits fluid leakage. Its bond strength is consistently lower than AH Plus and iRoot SP, attributed to weak tag-like structures (Assmann et al., 2012; Sagsen et al., 2011), and its microleakage exceeds MTA and MTA Plus (Sönmez et al., 2012). Most studies find it cytotoxic — possibly from lead release or its salicylate resin — with a reduction in cell viability (Bin et al., 2012; Zhou et al., 2015), although one report described recovery of viability after three

days and apatite formation in osteoblast-like cells (Salles et al., 2012). It shows antibiofilm activity against E. faecalis (Faria-Júnior et al., 2013).

3.3 MTA Plus

MTA Plus is a lower-cost, finer-powder material with a composition similar to white ProRoot MTA that can be mixed with either a gel or a liquid (Gandolfi et al., 2014). Mixed with gel it serves as a root-canal sealer with improved handling and washout resistance (DeLong et al., 2015); varying the powder-to-gel ratio tunes setting time and rheology (Gandolfi et al., 2014). It shows greater reactivity than ProRoot MTA, with sustained calcium release and alkalinisation linked to its solubility, fine powder, water sorption, porosity and calcium-phosphate formation. Both MTA Plus and BC Sealer bond well in a single-cone technique, but MTA Plus bonds less well under continuous-wave compaction, while showing minimal chemical change under vertical gutta-percha compaction (Viapiana et al., 2015). Its cytotoxic risk is minimal and concentration- and time-dependent, and the grey form may support cell viability better than white (Eid et al., 2014).

3.4 MTA Obtura

MTA Obtura resembles grey MTA — Portland-cement clinker and bismuth oxide — mixed with a liquid resin that also initiates cure (Vasconcelos et al., 2011). It combines MTA's biologic and sealing behaviour for gutta-percha or Resilon obturation (Camilleri et al., 2011). Its flow (27.65 mm) is lower than AH Plus (37.47 mm) but still exceeds the ADA/ISO minimum of 20 mm (Bernardes et al., 2010). It seals well to day 30, but leakage becomes severe by day 60 (Vasconcelos et al., 2011).

At a glance — bioceramic sealers

- Extremely hydrophilic: canal and dentinal-tubule moisture is an advantage, not a hazard.
- pH above 12 while unset gives calcium-hydroxide-like antibacterial activity.
- Slight expansion, no shrinkage, insolubility in tissue fluids, and bonding to bioceramic-coated gutta-percha eliminate the core–sealer gap.
- Long-term randomised trials are still needed to confirm clinical performance.

A systematic review of pre-mixed calcium-silicate sealers found good in-vitro physicochemical and biological properties and results in animal studies that matched or exceeded conventional sealers, while noting the shortage of well-designed long-term clinical trials (Almeida et al., 2017). Trope et al. (2015) summarised the class: highly hydrophilic, so canal moisture helps rather than harms; unset pH above 12 for antibacterial activity; setting at variable rates in different environments; slight expansion without shrinkage; insolubility in tissue fluids; and bonding to bioceramic-coated gutta-percha. Randomised double-blind trials remain necessary to confirm clinical performance.

CHAPTER FOUR

Clinical Uses of Bioceramic Materials

From the surgical root end to the immature tooth

The properties surveyed in the preceding chapters translate into a broad clinical repertoire. This chapter follows the material from the surgical root end, through perforation and resorption repair, to canal obturation, vital pulp therapy and the management of the immature tooth.

1. Retrograde filling

Endodontic surgery preserves teeth with post-treatment periapical disease when orthograde retreatment cannot address the cause — apical infection in inaccessible areas, foreign-body reactions, extraradicular infection or true cysts (Wu and Wesselink, 2005). Over two decades apical surgery has become microsurgery, adding magnification, illumination, microinstruments, ultrasonic contra-angle tips and more biocompatible materials (Kim and Kratchman, 2006). After the root end is resected, a retrograde filling seals the canal from the apical direction.

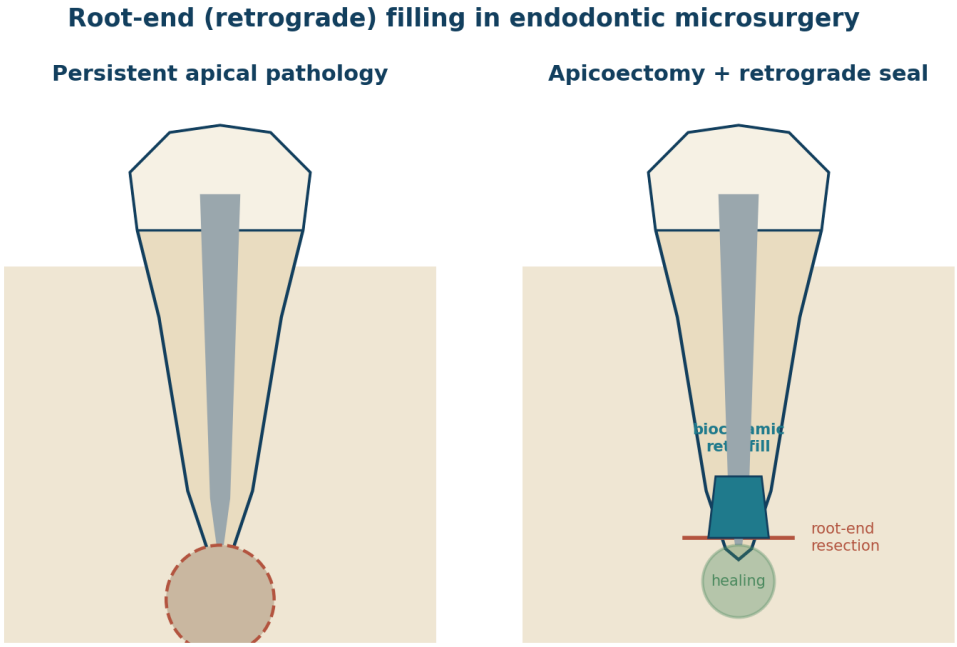


Figure 4.1 In apicoectomy the diseased root end is resected and a biocompatible bioceramic retrofill seals the canal, allowing periapical healing.

An ideal root-end material has demanding requirements (San Chong et al., 2004):

1. Bonds to dentinal walls and seals the root end in three dimensions.
2. Prevents rather than encourages the growth of pathogenic micro-organisms.
3. Dimensionally stable and impervious to moisture, set or unset.
4. Well tolerated by periradicular tissues, with no inflammatory reaction.
5. Able to stimulate regeneration of the periodontium.
6. Not cytotoxic locally or systemically.
7. Not electrochemically active and does not corrode.
8. Does not discolour tooth or periradicular tissues.
9. Radiopaque.
10. Easy to handle, with a long shelf life.

Table 4. Requirements of an ideal root-end filling material.

Because materials available before 1993 met few of these criteria, MTA was introduced specifically as a root-end filling and quickly gained a reputation for superior sealing and biocompatibility (Kim and Kratchman, 2006; Torabinejad et al., 1995b). In a classic bacterial-leakage study, most amalgam, Super-EBA and IRM samples leaked within 6–57 days, whereas MTA resisted leakage across 90 days (Torabinejad et al., 1995c); MTA also consistently promoted regeneration of periodontal ligament and cementum (Kim and Kratchman, 2006). MTA Plus proved equivalent to MTA as a root-end filling at one-year follow-up (Zhou et al., 2017).

MTA's limitations — difficult handling, long set and low washout resistance — spurred development of newer root-repair bioceramics combining calcium silicate and calcium phosphate: BioAggregate, ERRM (putty and paste) and iRoot FS. BioAggregate matched white MTA for leakage as a root-end filling (Leal et al., 2011) and stimulated odontoblastic differentiation and mineralisation in pulp cells (Zhang et al., 2013). In root-end microsurgery in dogs, ERRM Putty produced better tissue healing at the resected surface than MTA (Chen et al., 2015), and the two had similar sealing ability in a nutrient-leakage model (Antunes et al., 2016).

2. Perforation repair

2.1 Furcal perforations

Furcation perforations connect the root-canal space to the periradicular tissues, arising from iatrogenic error or from pathology such as caries and resorption (AAE, 2015). The resulting inflammation damages the periodontal ligament and resorbs bone and dental tissue, and can rapidly open a route to the gingival sulcus with irreversible loss of attachment.

Sites of iatrogenic root perforation

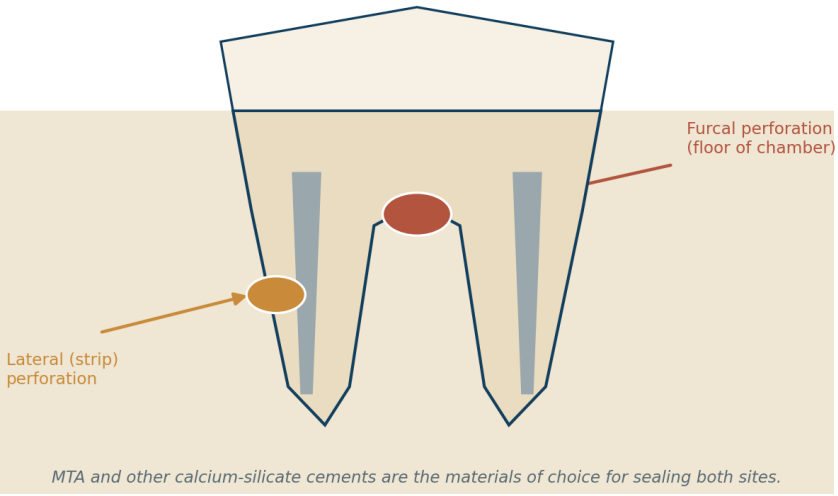


Figure 4.2 Perforations at the furcal floor and along the lateral root wall both demand an immediate, biocompatible, moisture-tolerant seal.

MTA remains the gold standard for furcal repair, giving a favourable periradicular response and induced mineralisation in vivo (Parirokh and Torabinejad, 2010b; Zairi et al., 2012). Because MTA's iron oxide can discolour marginal gingiva and crown, tricalcium-silicate cements such as Biodentine — without staining heavy metals, with good sealing, antimicrobial activity and a shorter set — were developed as alternatives (Kang et al., 2015). Biodentine allowed formation of mineralised bridges thicker than those with MTA after pulpotomy in dogs (De Rossi et al., 2014); a later study found MTA gave a higher rate of complete sealing and thicker new mineralised tissue, with Biodentine still adequate (Silva

et al., 2017). Both materials induce mineralisation markers — alkaline phosphatase, osteopontin and RUNX2 (Daltoé et al., 2016).

2.2 Lateral root perforations

Lateral perforations can occur at any canal level; the more coronal the site, the easier it is to repair the perforation and obturate the canal separately. The canal is usually filled first, carefully avoiding sealer contamination of the perforation, and during repair the cement is kept from dropping into the empty canal. Success depends on the elapsed time before treatment and the perforation's location and size (Fuss and Trope, 1996). Ford et al. (1995) found no clear link between size and prognosis, and MTA has healed large perforations in case reports (Arens and Torabinejad, 1996).

Location is decisive. Fuss and Trope (1996) defined the critical zone at the crestal bone and epithelial attachment: perforations apical or coronal to it have a good prognosis, whereas crestal perforations invite epithelial migration and periodontal breakdown and heal least well; a periodontal defect connecting the mouth to the perforation worsens the outlook (Ford et al., 1995). Treatment should be immediate sealing with a biocompatible material that is insoluble in tissue fluids and promotes tissue regeneration (Beavers et al., 1986; Holland et al., 2007). Older materials — amalgam, Cavit, calcium hydroxide, Super-EBA, calcium-phosphate cement, glass ionomer, decalcified freeze-dried bone — are insufficiently biocompatible against bone, so teeth with perforations carried an uncertain prognosis before MTA (Farzaneh et al., 2004; Fuss and Trope, 1996). MTA has since proved successful for both furcal and lateral perforations (Main et al., 2004; Regan et al., 2002).

Lemon (1992) described the requirements of an ideal repair material (Table 5):

1. Allows a hermetic seal.
2. Insoluble and non-resorbable.
3. Impervious to haemorrhage.
4. Manageable, to avoid extrusion.
5. Biocompatible.

6. Stimulates osteogenesis and cementogenesis.
7. Non-toxic and non-carcinogenic.
8. Available and easily obtainable.
9. Not harmful to patient or dentist.
10. Inexpensive.

Table 5. Requirements of an ideal repair material.

3. Internal inflammatory root resorption

3.1 Internal resorption in the root canal

Internal resorption follows destruction of the organic sheath, predentine and odontoblast layer, exposing mineralised dentine to pulpal cells with resorptive potential. Contrary to earlier belief it is often associated with pulpitis and stops once the pulp becomes necrotic (Gabor et al., 2012). Predisposing factors include trauma, pulpotomy, transplantation, invagination, restorative and orthodontic procedures, and even herpes-zoster infection. It is rarer than external resorption and can be confused with external cervical resorption, risking misdiagnosis (Frank, 1981; Patel et al., 2009).

Once confirmed and if the tooth is restorable, conventional root-canal treatment is performed to remove vital and necrotic remnants that could feed the resorbing cells and to disinfect and fill the system (Endodontology, 2006). Where the root is not perforated, chemo-mechanical debridement with calcium hydroxide (for large defects) followed by gutta-percha and sealer is sufficient — bioceramics are not specifically required (Haapasalo et al., 2015). Once resorption has perforated the root, however, bioceramic cements are an excellent choice: they tolerate moisture and small amounts of blood, seal well to dentine, are antimicrobial and have dentine-like mechanical properties (Altundasar and Demir, 2009; Torabinejad et al., 1995a).

3.2 Internal resorption in the pulp chamber

Non-perforating chamber resorption does not require bioceramics and is straightforward to manage. If a perforation has occurred, the material is chosen by size and location: bioceramic cements are tapped gently

against the tissue with large inverted paper points to fill perforations apical to the junctional epithelium, while the rest of the chamber is restored according to the overall plan (Haapasalo et al., 2015).

4. Invasive cervical resorption

4.1 Type I and II cervical resorption

Invasive (external) cervical resorption (ECR) begins just below the epithelial attachment and remains poorly understood. Defects in the cementum layer expose dentine to clastic cells, and the anatomy of the cemento-enamel junction — where cemental gaps are common — leaves dentine unprotected and liable to osteoclastic resorption (Fuss et al., 2003; Neuvald and Consolaro, 2000). Predisposing factors include trauma, orthodontics, internal bleaching, surgery, periodontal therapy, bruxism, intracoronal restorations, developmental defects and systemic disease (Heithersay, 1999; Patel et al., 2009). ECR is usually aseptic but can be secondarily colonised by micro-organisms from the sulcus or a necrotic pulp (Heithersay, 2004; Trope, 2002).

Heithersay (2004) classified ECR by depth of invasion: Class 1, shallow and cervical; Class 2, into coronal dentine near the pulp; Class 3, into the coronal third of the root; Class 4, beyond the coronal third. Clinically it presents in the cervical region, sometimes as a pink spot, with a positive vitality response until the pulp is involved; advanced lesions bleed profusely on probing and have sharp cavity margins (Patel et al., 2009).

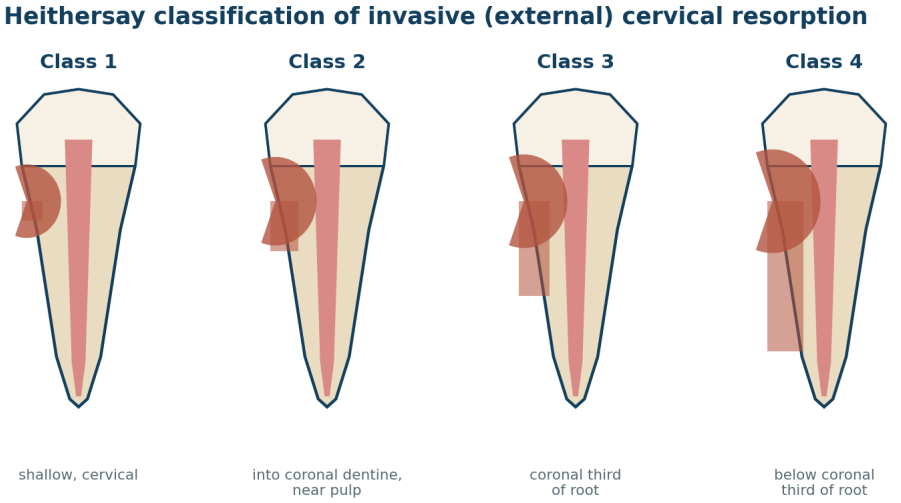


Figure 4.3 The Heithersay classes describe progressive apical and pulpal invasion of the resorptive lesion; class guides both access and restorative material.

Treatment aims to remove the resorptive soft tissue, separate the lesion from its periodontal attachment to stop clastic-cell repopulation, and restore the cavity — dictated by severity, location, any perforation and restorability (Patel et al., 2009). Class 1 and 2 lesions are often treated surgically from outside with reinforced glass ionomer or bonded composite; bioceramic cements may partly wash out during the slower set (Haapasalo et al., 2015). If the cavity is close to the pulp, root-canal treatment precedes flap reflection, lesion removal and curettage of granulomatous tissue to cut the clastic-cell blood supply, before restoration and suturing (Trope, 2002). A deep Class 2 lesion may instead be treated through the canal (Haapasalo et al., 2015). Heithersay (2004) advised topical 90% trichloroacetic acid to induce coagulation necrosis of the lesion before curettage and glass-ionomer restoration.

4.2 Type III and IV cervical resorption

Most Class 3 and 4 lesions are accessed through the canal without surgery, making bioceramic cements very suitable for restoring the resorptive region (Bakland and Andreasen, 2012; Park and Lee, 2008). Inter-appointment calcium hydroxide is used after access, irrigation and removal of resorption-affected dentine; the canal apical to the resorption is filled with gutta-percha and sealer or with bioceramic cement, and the large resorptive region is filled with bioceramic cement placed via a

carrier and large paper point. A cotton pellet over the cement until the next visit confirms complete setting. RRM Putty and FS Putty aid radiographic assessment through high radiopacity, whereas Biodentine is less radiopaque (Haapasalo et al., 2015).

Because of their mechanical and biological behaviour, bioceramic cements can be used across the stages of invasive cervical resorption. Periodontal reattachment is unlikely with bonded composite, amalgam or glass ionomer, but experimental evidence suggests it may be feasible with MTA in this position (Ford et al., 1995; Koh et al., 1997).

5. Orthograde bioceramic root-canal filling

Obturation provides a fluid-tight barrier that protects the periradicular tissues from oral micro-organisms. Three goals define a sound obturation: prevent coronal leakage of organisms and nutrients, prevent periodontal and periapical fluids from leaching in to feed organisms, and entomb any organisms surviving disinfection (Sundqvist and Figdor, 1998). Grossman (1978) listed the ideal properties of a root-filling material (Table 6):

1. Easy to handle, with adequate working time.
2. Seals the canal laterally and apically and adapts to complex anatomy.
3. Dimensionally stable.
4. Non-irritant to periapical tissues.
5. Does not stain tooth structure.
6. Antimicrobial.
7. Impervious to moisture and non-porous.
8. Unaffected by tissue fluid.
9. Radiopaque.
10. Easily removed from the canal.

Table 6. Properties of an ideal root-canal filling.

No material meets every requirement, and an added ideal — active stimulation of periradicular regeneration — favours bioceramics: MTA regenerated human periodontal ligament with cement-like tissue on its surface, matching animal findings (Bartols et al., 2017). Because gutta-percha and sealer can leak within 3–30 days in vitro (Fransen et al., 2008; Madison and Wilcox, 1988), the canal or its apical part is sometimes obturated with a bioceramic cement (Al-Hezaimi et al., 2005; De-Deus et al., 2008): before a planned apicoectomy, to simplify surgery; where internal resorption has perforated the mid or apical canal; or in persisting symptoms, where the sealing ability and antimicrobial activity of the cement may resolve the case (Haapasalo et al., 2015). Bioceramic cements are chosen when the material–tissue interface is larger than normal — wide open apices, perforations and retrograde fillings (Tomson et al., 2014) — and MTA is an option for complete or partial filling of anatomically complex teeth such as C-shaped canals, fusion/gemination and dens evaginatus (Bogen and Kuttler, 2009).

6. Pulp capping and vital pulp therapy

Preserving pulp vitality is vital in immature permanent teeth, where continued vitality allows complete root formation (Laurent et al., 2012). Vital pulp therapy aims to retain the pulp and induce the pulp–dentine complex to regenerate and form tertiary dentine — especially important where the apex is still developing (Bogen and Chandler, 2008). Options range from indirect and direct pulp capping to partial and full pulpotomy, chosen by the extent of pulp involvement.

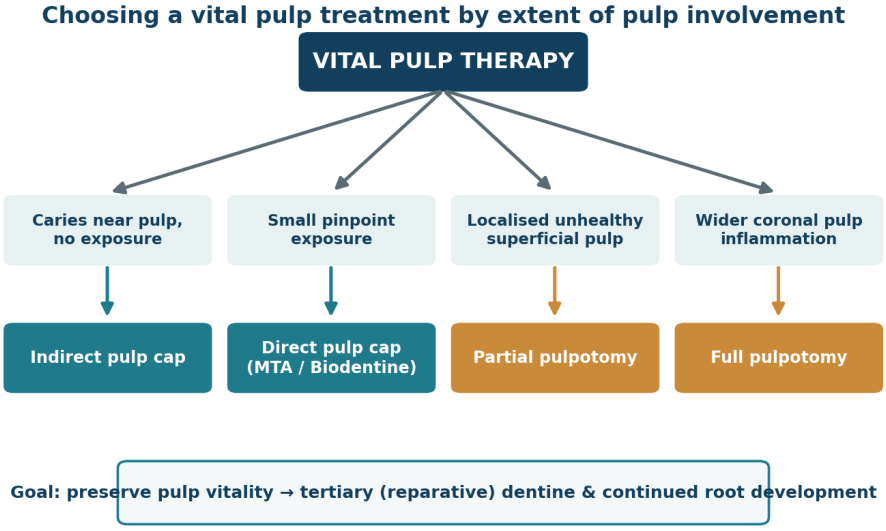


Figure 4.4 A practical way to picture vital pulp therapy: as pulp involvement increases, the intervention escalates from indirect capping to full pulpotomy.

The American Association of Endodontists defines direct pulp capping (DPC) as sealing an exposed vital pulp with a material such as calcium hydroxide or MTA to promote reparative dentine and maintain vitality (AAE, 2003). The ideal capping material seals the wound, differentiates pulp cells into odontoblasts, is biocompatible and antimicrobial, and has adequate mechanical strength (Koh et al., 1997; Zhang et al., 2013). Calcium hydroxide was traditional but has drawbacks — tunnel defects in the dentine bridge, high solubility, chamber obliteration, poor adhesion and degradation after acid etching (Asgary and Ahmadyar, 2013; Stanley and Pameijer, 1997) — so MTA and Biodentine are now preferred.

MTA induces dentinogenesis more quickly and homogeneously than calcium hydroxide and encourages granulation-tissue formation that aids healing (Dammaschke et al., 2010); bioceramics also surpass calcium hydroxide in mechanical properties, sealing and often biocompatibility (Ma et al., 2011). A randomised trial of 376 patients confirmed MTA's superiority as a DPC material at two years (Hilton et al., 2013), and in carious molar exposures MTA outperformed calcium hydroxide (85% vs 52% success; Kundzina et al., 2016). MTA and Biodentine each achieved 100% success in young first permanent molars over 6–12 months (Katge

and Patil, 2017), and iRoot BP Plus matched MTA for complete barrier formation in dogs (Shi et al., 2016).

Partial pulpotomy removes only superficial unhealthy pulp, preserving the bulk of vital tissue (Cvek, 1978). ProRoot MTA and hard-setting calcium hydroxide both gave favourable outcomes over two years, with worse results when exposures exceeded 5 mm^2 (Chailertvanitkul et al., 2014). Although classically contraindicated in irreversible pulpitis, MTA performed well in mature teeth: partial pulpotomy succeeded in 85% of MTA cases versus 43% for calcium hydroxide at two years (Taha and Khazali, 2017). Biodentine and ProRoot MTA induced complete dentinal bridges and outperformed TheraCal (Bakhtiar et al., 2017). A full pulpotomy removes the entire coronal pulp to the canal orifices and shares the indications of partial pulpotomy where inflammation is more extensive (Trope, 2008). Grey MTA has been advocated for pulpotomy even in permanent teeth with irreversible pulpitis (Barrieshi-Nusair and Qudeimat, 2006), and in a retrospective series MTA succeeded in 84% of irreversible-pulpitis teeth and 76% of teeth with periapical radiolucency (Linsuwanont et al., 2017). Many cements appear broadly interchangeable — Biodentine matched MTA for mineralised-bridge formation (De Rossi et al., 2014).

7. Teeth with necrotic pulps and open apices

A young permanent tooth with pulp necrosis is a restorative and endodontic challenge: loss of vitality halts dentine and apex formation, leaving thin, fracture-prone walls — the immature tooth. Apexification induces a calcified apical barrier or continued apical development (AAE, 2003).

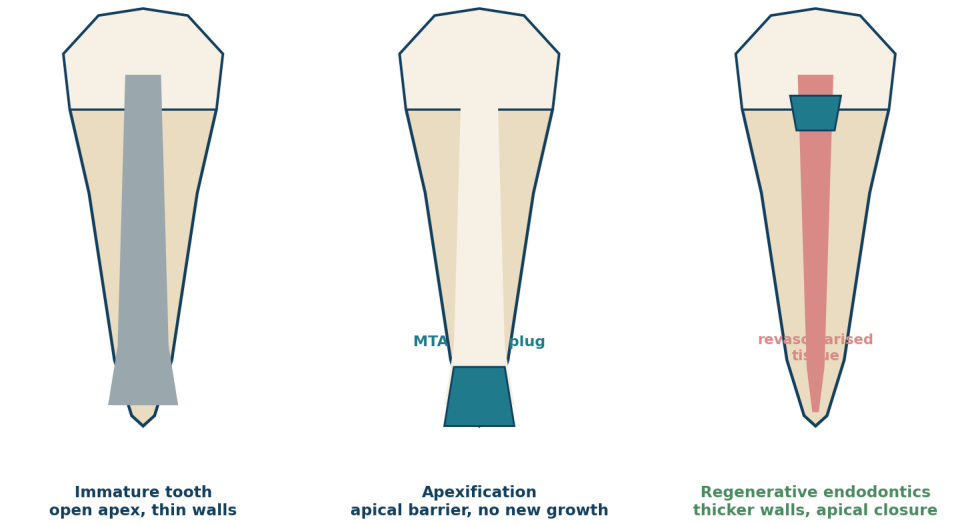


Figure 4.5 Apexification builds an artificial apical barrier; regenerative endodontics aims to thicken the walls and close the apex with new vital tissue.

Calcium hydroxide apexification is effective (success up to ~90%) but slow and unpredictable, needs multiple visits, risks reinfection and fracture, and leaves a porous 'Swiss-cheese' barrier (Ballesio et al., 2006; Felippe et al., 2005). Bioceramic cements are now the preferred option: a plug is tapped into the apical canal at the first or second visit and the remaining canal filled as the case requires (Hachmeister et al., 2002; Karp et al., 2006; Mente et al., 2013). MTA gives earlier radicular and coronal filling and reduces fracture risk relative to calcium hydroxide (Mente et al., 2013); in one outcome study all MTA cases healed versus 2 of 15 calcium-hydroxide failures (El Meligy and Avery, 2006), and MTA reached comparable healing in significantly less treatment time (Pradhan et al., 2006).

MTA is widely used for single-visit apexification for its regenerative capacity, antibacterial action, biocompatibility, sealing and improvement of fracture resistance (Yadav et al., 2015). Despite handling difficulty, long set and discolouration, 96% of non-vital immature teeth treated with an MTA barrier and bonded restoration were healed and functional at 5–15 years (mean 8.29; Ree and Schwartz, 2017). Biodentine has shown favourable outcomes as an apical plug (Vidal et al., 2016);

micro-CT found ProRoot MTA better adapted than MTA Angelus, with mechanically mixed products handling better (Sisli and Ozbas, 2017), while ProRoot MTA, MTA Plus and BC Root Repair Material-FS Putty adapted similarly, the last improved by prior BC Sealer application (Tran et al., 2016).

8. Regenerative endodontics

Immature necrotic teeth may be managed by long-term calcium-hydroxide apexification, MTA apexification, or regenerative endodontic treatment (RET) (Hargreaves et al., 2013). Long-term calcium hydroxide weakens root structure, and although an MTA plug allows earlier filling, MTA apexification does not thicken the walls or promote root development, so both apexification methods leave the tooth fracture-prone (Cvek, 1992; Jeeruphan et al., 2012; Mente et al., 2013; Torabinejad et al., 2017). RET, first proposed in the 1970s by Nygaard-Østby, seeks to overcome this.

RET comprises 'biologically based procedures designed to replace damaged structures, including dentine and root structures and cells of the pulp–dentine complex' (Murray et al., 2007). Torabinejad and McDonald (2009) described the ideal RET material (Table 7):

1. Withstands bacterial leakage long-term.
2. Not difficult to handle.
3. Radiopaque.
4. Impervious to moisture.
5. Non-absorbable in tissue fluids.
6. Non-toxic to human tissue.
7. Non-carcinogenic.
8. Does not stain tooth structure.
9. Impenetrable.
10. Antibacterial and antifungal.
11. Biocompatible.

Table 7. Properties of an ideal material for RET.

It should also help the residual pulp recover and encourage new hard and soft tissue — requirements largely met by bioceramics. White MTA is the material of choice, placed as a plug in the middle or coronal third after disinfection with triple antibiotic paste; a blood clot or platelet-rich plasma provides an apical scaffold before MTA seals the access (Diogenes et al., 2013). MTA use may support periodontal-ligament healing (Maeda et al., 2010), and Biodentine and BioAggregate matched MTA in stimulating odontoblastic differentiation and mineralisation via MAP-kinase activation (Jung et al., 2015).

Retrospectively, RET produced greater increases in root length and thickness than MTA or calcium-hydroxide apexification, with survival of 100%, 95% and 77.2% respectively (Jeeruphan et al., 2012). Yet RET and MTA apexification gave comparably good disease resolution and tooth retention without predictable further root development (Silujjai and Linsuwanont, 2017), and high-quality head-to-head trials are still lacking — large, long-term randomised multicentre studies are needed (Torabinejad et al., 2017).

CHAPTER FIVE

Bioceramics across Dentistry

Restorative, prosthodontic, implant and regenerative uses

About this chapter

This chapter is expository background compiled for the expanded edition. It summarises widely accepted dental-materials science and, unlike the endodontic chapters, does not carry author–year citations. Product names are given only as familiar examples.

The endodontic materials described so far are one branch of a much larger family. The same ceramic chemistry — and the same idea of a material that is either inert, bioactive or resorbable — runs through almost every discipline of dentistry, from the crown that restores a tooth to the graft that rebuilds the bone beneath it. This chapter surveys that wider landscape and shows how the properties emphasised earlier (biocompatibility, apatite formation, alkalinity, strength) reappear in new clinical roles.

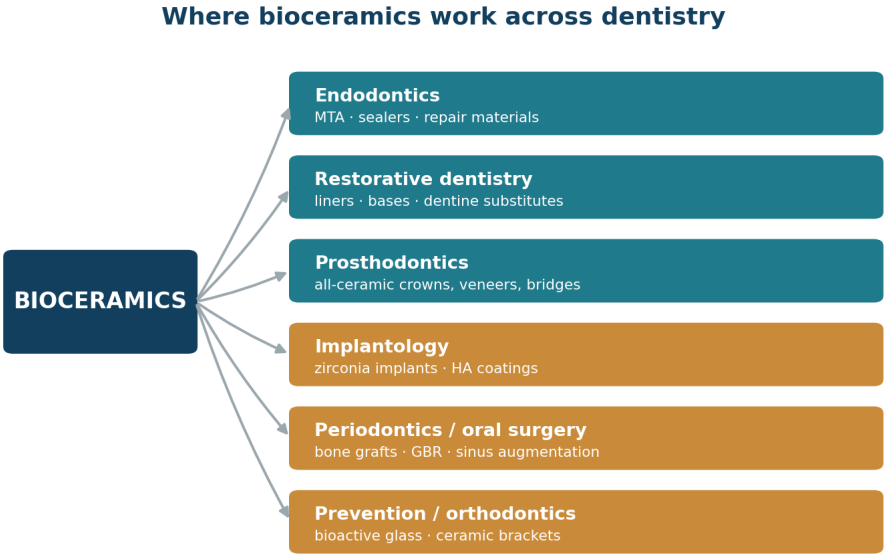


Figure 5.1 A single materials family serves many dental disciplines; the calcium-silicate cements of endodontics sit alongside structural ceramics, implant materials and regenerative scaffolds.

5.1 A common chemistry, many roles

It is useful to separate two broad groups. Structural (restorative) ceramics — dental porcelains and oxide ceramics — are prized chiefly for

strength, wear resistance and appearance; most are essentially bioinert and are chosen to replace lost tooth structure above the gum line. Bioactive and resorbable bioceramics — calcium silicates, hydroxyapatite, other calcium phosphates and bioactive glasses — are chosen for how they interact with living tissue: they seal, they release ions, they template new mineral and, in some cases, they are replaced by the tissue they help to heal. Endodontic bioceramics belong firmly to the second group, but a complete picture of dentistry requires both.

5.2 Restorative dentistry: liners, bases and dentine substitutes

Beneath a filling, the pulp must be protected from thermal, chemical and bacterial insult. For decades calcium hydroxide was the standard liner over deep dentine and near exposures, prized for its alkalinity and its ability to stimulate reparative dentine — but limited by solubility, low strength and a porous dentine bridge. Calcium-silicate materials have changed this picture. The same tricalcium-silicate chemistry that makes MTA valuable in the root canal makes it, and its faster-setting relatives, excellent dentine substitutes.

Biodentine is marketed explicitly as a dentine-replacement material: it can line or base a deep cavity, cap an exposed pulp, and even serve as a temporary full-dentine restoration under a definitive coronal restoration, thanks to its short set, useful strength and calcium release. Light-cured calcium-silicate liners add resin to allow on-command setting and easy placement as a thin protective layer under composite. In each case the attraction is bioactivity — an alkaline, calcium-releasing surface that encourages mineralisation — combined with better mechanical behaviour and lower solubility than calcium hydroxide. Conventional and resin-modified glass-ionomer cements, while not calcium-silicate ceramics, occupy an adjacent niche: they bond to tooth structure, release fluoride and are often described as bioactive, and they remain widely used as liners, bases and restorations.

At a glance — the modern liner/base

- Calcium-silicate liners give an alkaline, calcium-releasing, bioactive surface over deep dentine.
- They set faster and are stronger and less soluble than traditional calcium hydroxide.
- Biodentine can act as a bulk dentine substitute beneath a definitive restoration.

5.3 Dental ceramics for restorations

When a tooth needs a veneer, inlay, onlay, crown or bridge, the material of choice is increasingly an all-ceramic restoration. Dental ceramics are conventionally grouped by microstructure: predominantly glassy ceramics (feldspathic porcelain), particle-filled glasses (leucite-reinforced and lithium-disilicate glass-ceramics), interpenetrating glass-infiltrated ceramics, and fully crystalline polycrystalline oxide ceramics (alumina and, above all, zirconia).

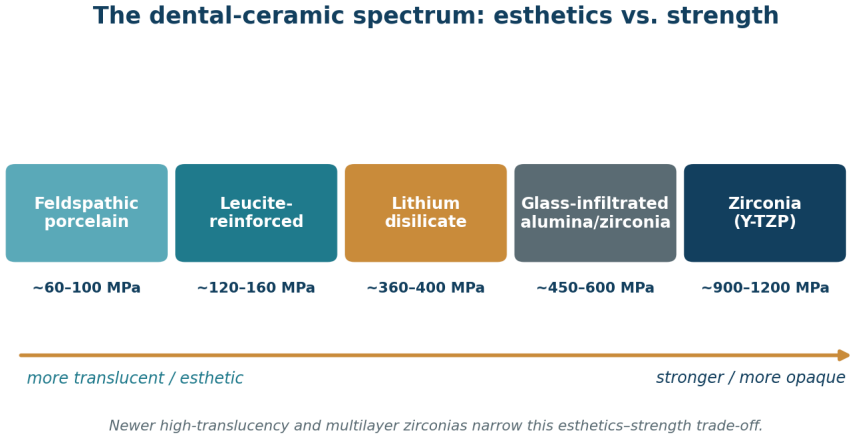


Figure 5.2 As microstructure shifts from glassy to polycrystalline, strength rises while translucency tends to fall — though modern high-translucency zirconias soften the trade-off.

The clinically important consequence is a trade-off between strength and appearance. Glassy feldspathic porcelain is exquisitely lifelike but relatively weak, suiting veneers and conservative anterior work. Leucite-

and lithium-disilicate glass-ceramics add strength while retaining good aesthetics, making lithium disilicate a workhorse for anterior and posterior crowns. Zirconia — yttria-stabilised tetragonal zirconia polycrystal (Y-TZP) — is the strongest, its toughness enhanced by a remarkable mechanism called transformation toughening, in which crystal grains change phase at an advancing crack tip and squeeze it shut. Zirconia is used for high-load crowns, multi-unit bridges, implant abutments and, increasingly, as monolithic (single-material) restorations; newer multilayer and high-translucency formulations have narrowed its historic aesthetic disadvantage.

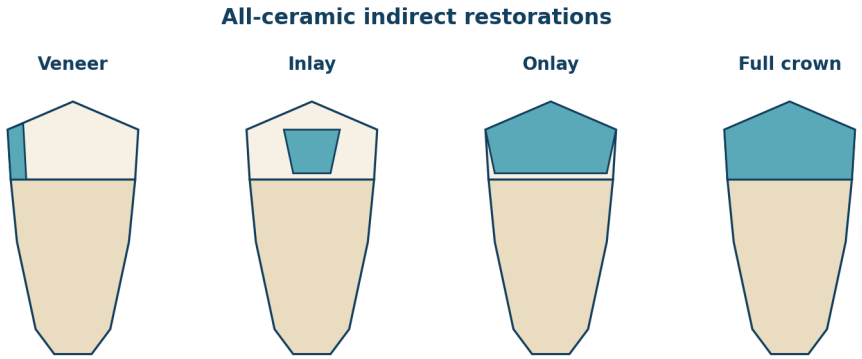


Figure 5.3 All-ceramic restorations range from conservative veneers, inlays and onlays to full-coverage crowns.

Most modern ceramic restorations are produced by CAD/CAM: a solid ceramic block is milled to shape from a digital design and, for oxide ceramics, sintered to full density. Because glassy and glass-ceramic materials can be etched and silane-treated, they bond predictably to tooth structure with resin cements; the denser oxide ceramics rely more on mechanical retention and specialised primers. The choice among ceramics is therefore a balance of load, aesthetic demand, tooth preparation and the bonding protocol available.

5.4 Prosthodontics and the restored tooth

Seen together, bioceramics appear at almost every level of a restored tooth. A ceramic crown or veneer rebuilds the visible surface; a calcium-silicate liner or base protects the pulp; and, where the tooth has been treated endodontically, a bioceramic cement or sealer occupies the

canal. In an edentulous site, a ceramic-surfaced implant carries a ceramic abutment and crown. The materials differ in purpose — structural above, bioactive below — but they share a common ceramic ancestry.

Bioceramics in the restored dentition

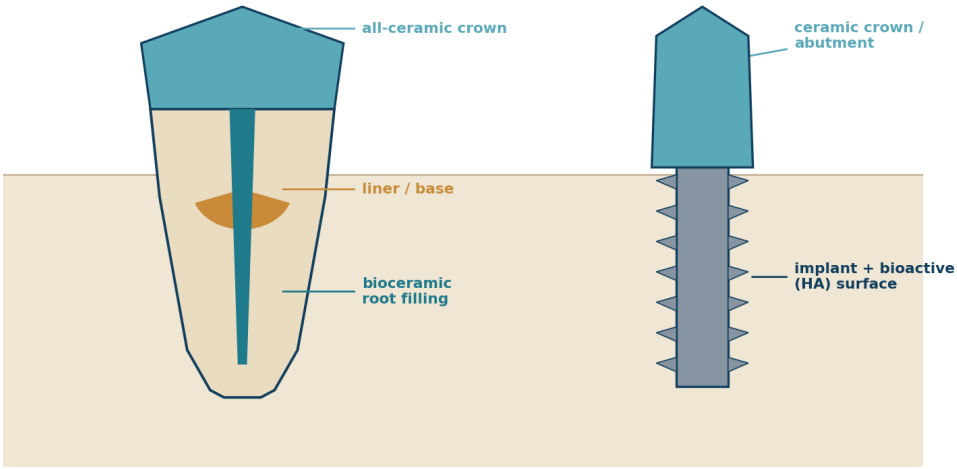


Figure 5.4 From crown to canal to implant, ceramic materials of different kinds cooperate to restore form and function.

5.5 Implant dentistry

Dental implants succeed through osseointegration — a direct structural and functional connection between living bone and the implant surface. While titanium remains the reference material, ceramics contribute in two ways. First, zirconia implants offer a metal-free, tooth-coloured alternative with good biocompatibility and soft-tissue response, attractive where aesthetics or metal sensitivity are concerns. Second, bioactive coatings — most familiarly plasma-sprayed hydroxyapatite — are applied to implant surfaces to speed and strengthen the bone response: the coating presents a calcium-phosphate surface onto which bone grows and bonds directly, rather than merely apposing an inert metal.

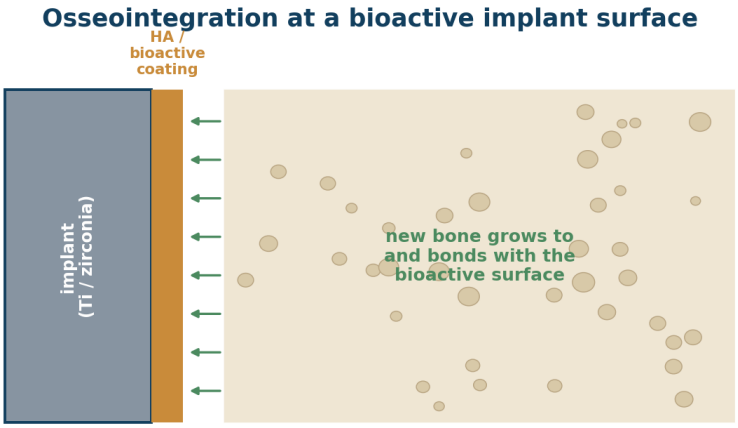


Figure 5.5 A bioactive (hydroxyapatite) surface invites bone to grow up to and bond with the implant, reinforcing osseointegration.

5.6 Bone regeneration: grafting and guided regeneration

Rebuilding lost alveolar bone — after extraction, around implants, in periodontal defects or during sinus augmentation — is one of the largest uses of bioceramics in dentistry. Calcium-phosphate ceramics such as hydroxyapatite and β-tricalcium phosphate (β-TCP), often combined as biphasic calcium phosphate, act as osteoconductive scaffolds: a biocompatible framework along which the patient's own bone-forming cells migrate and lay down new bone. Hydroxyapatite is relatively stable and long-lasting; β-TCP resorbs more readily and is progressively replaced by host bone; biphasic materials are engineered to tune this balance.

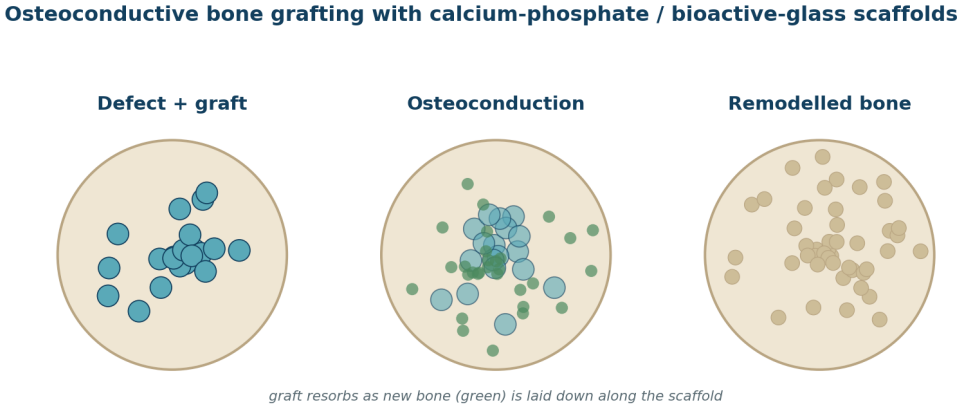


Figure 5.6 An osteoconductive graft provides a scaffold that is gradually resorbed and replaced by the patient's own remodelled bone.

It helps to distinguish three related ideas. Osteoconduction is the passive provision of a scaffold for bone ingrowth — the principal action of calcium-phosphate ceramics. Osteoinduction is the active recruitment and differentiation of progenitor cells into bone-forming cells, typically driven by signalling molecules. Osteogenesis is the formation of bone by living cells within a graft. Synthetic ceramic grafts are chiefly osteoconductive, which is why they are frequently combined with barrier membranes in guided bone regeneration — the membrane excludes fast-growing soft tissue so that slower bone can fill the ceramic scaffold undisturbed.

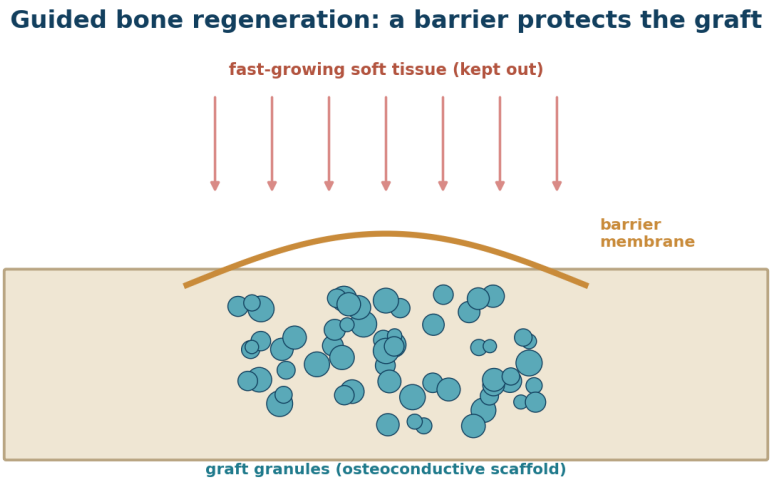


Figure 5.7 In guided bone regeneration a barrier membrane holds back fast-growing soft tissue, giving the slower bone time to fill the osteoconductive graft.

5.7 Bioactive glass

Bioactive glass — the material with which Larry Hench opened the modern bioceramics era — deserves its own mention because it appears in several dental roles. Its defining trick is a sequence of surface reactions in body fluid: ions exchange out of the glass, a hydrated silica-gel layer forms, calcium and phosphate accumulate on it, and a crystalline hydroxy-carbonate-apatite layer develops that bonds directly to tissue — the very chemistry that underlies the bioactivity of endodontic calcium silicates.

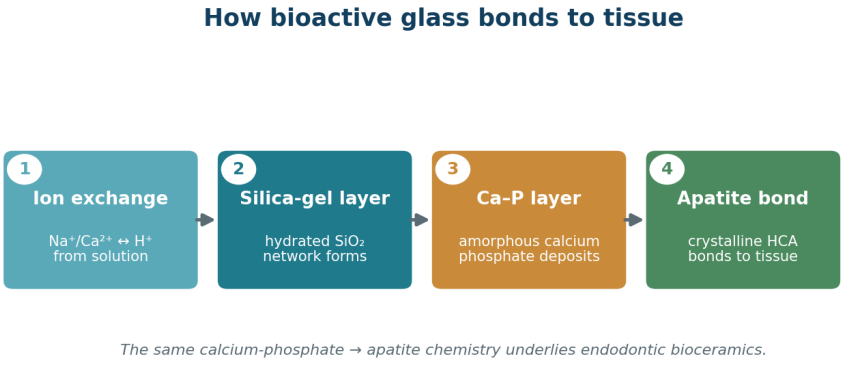


Figure 5.8 The bioactive-glass surface reaction ends in an apatite layer that bonds to tissue — the same end-point as calcium-silicate bioceramics.

In dentistry this makes bioactive glass useful as a bone-grafting material, as a component of remineralising and desensitising agents (calcium-sodium-phosphosilicate particles in toothpastes and prophylaxis pastes occlude dentinal tubules and release mineral-forming ions), and as an air-abrasion powder. The common thread is the material's ability to deliver calcium and phosphate to a mineralising surface.

5.8 Orthodontics, periodontics and maxillofacial use

Ceramics appear in several further niches. Ceramic orthodontic brackets made of alumina or zirconia offer a tooth-coloured, aesthetic alternative to metal. In periodontics, the calcium-phosphate and bioactive-glass grafts already described are used to regenerate intrabony and furcation defects. In oral and maxillofacial surgery, alumina, hydroxyapatite and related ceramics have long been used to reconstruct and augment bone, echoing their orthopaedic origins. Across all of these, the selection principle is the same one that governs endodontic practice: match the material's inertness, bioactivity or resorbability to the biological task at hand.

5.9 Prevention and emerging directions

The frontier of dental bioceramics is increasingly biological and additive. Remineralising formulations aim to reverse early caries and dentine

hypersensitivity by delivering calcium and phosphate to the tooth surface. Antibacterial bioceramics exploit the high pH and ion release already familiar from endodontic materials. Additive manufacturing (3D printing) of ceramics promises patient-specific scaffolds and restorations, and tissue-engineering approaches combine resorbable ceramic scaffolds with cells and signalling molecules to regenerate — rather than merely replace — dental and bony tissue. Regenerative endodontics, discussed in Chapter Four, is one clinical expression of this same ambition.

Chapter summary

- Structural ceramics (porcelains, zirconia) restore the visible tooth; bioactive/resorbable bioceramics interact with tissue below it.
- Calcium-silicate liners and dentine substitutes bring endodontic bioactivity into restorative dentistry.
- Hydroxyapatite, β-TCP and bioactive glass rebuild bone by osteoconduction, often with guided regeneration.
- Bioactive coatings strengthen implant osseointegration; bioactive glass also remineralises and desensitises teeth.
- The shared end-point — apatite formation and tissue bonding — links every branch of the family.

CHAPTER SIX

Understanding Bioceramic Materials

Properties, testing and handling

About this chapter

Like Chapter Five, this is background material for the expanded edition, written without new citations. It gathers, in one place, the properties and test methods referred to throughout the endodontic chapters.

The clinical behaviour of a bioceramic follows directly from a handful of measurable properties. Understanding how each is defined and tested makes the comparisons in earlier chapters easier to interpret and helps the clinician read a manufacturer’s data sheet critically.

6.1 Setting chemistry

Calcium-silicate bioceramics set by hydration, not by drying: water reacts with the silicate phases to form calcium-silicate-hydrate gel and calcium hydroxide (see Figure 2.1). The calcium hydroxide dissociates to raise pH and supply calcium ions, while the hydrate gel provides structure. Because water drives the reaction, these materials tolerate — indeed require — the moist environment that degrades many conventional cements, a property that recurs in every clinical application from root-end filling to implant coatings.

6.2 How key properties are measured

Radiopacity lets the clinician confirm a filling radiographically. It is measured against an aluminium step-wedge and expressed as an equivalent thickness of aluminium; international standards set a minimum of about 3 mm. Radiopacifiers such as bismuth oxide or zirconium oxide are added to reach clinically useful values (compare Figure 2.4).

Setting time is determined with standardised indenter (Gilmore or Vicat) needles, recording the initial set (when the surface first resists indentation) and the final set. For calcium-silicate materials the result depends strongly on water availability, particle size and any accelerators, which is why products range from a few minutes to several hours (Figure 2.2).

Solubility and dimensional stability matter because a material that dissolves or shrinks loses its seal. Solubility is assessed by mass loss after immersion; a good root-end or repair material shows very low solubility and slight expansion rather than shrinkage. Flow, important for sealers, is measured by compressing a fixed volume between plates and recording the disc diameter, against a standardised minimum.

Mechanical properties are characterised by compressive and flexural strength, microhardness, and — specific to endodontics — push-out bond strength, which measures resistance to displacement from the canal wall. These values are sensitive to mixing, porosity, environmental pH and time, so laboratory figures should be read as comparative rather than absolute.

pH and calcium-ion release are followed over time in solution. The characteristic rise to a strongly alkaline pH (Figure 2.3) reflects continuing calcium-hydroxide formation and is responsible both for antibacterial activity while the material is unset and for the apatite formation that follows.

6.3 Biological evaluation

Biocompatibility is not a single measurement but a hierarchy of evidence. Cytotoxicity assays (for example the MTT and XTT tests) expose cultured cells to the material or its extracts and measure survival; cell-attachment and morphology studies assess how well cells settle and spread on the set surface. Biomineralisation markers — alkaline phosphatase, osteopontin, osteonectin and transcription factors such as RUNX2 — indicate whether a material is actively driving cells toward a hard-tissue-forming phenotype. Above these sit animal studies and, at the top, randomised controlled clinical trials. As the endodontic chapters repeatedly note, the laboratory and animal evidence for bioceramics is strong, while long-term randomised clinical data remain the principal gap.

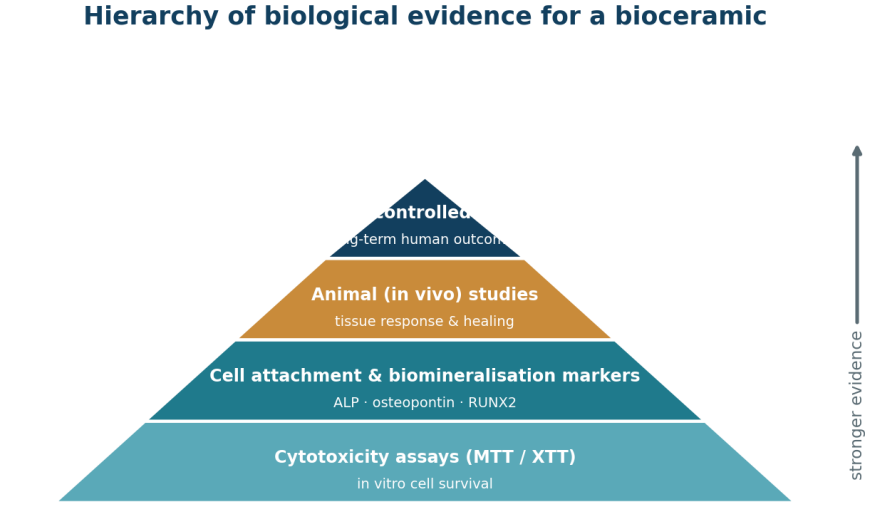


Figure 6.1 Confidence in a material grows from laboratory assays at the base to randomised clinical trials at the apex; for bioceramics the base is well populated and the apex remains the main gap.

6.4 Handling and clinical tips

Several practical points recur across applications. Mix to the manufacturer's ratio and avoid incorporating air, which raises porosity and lowers strength. Provide adequate — but not excessive — moisture: too little prevents set, too much can prolong it and soften the surface. Place the material without voids and, where a slow-setting cement is used, protect it (for example with a moist cotton pellet) until set is confirmed. Where discolouration matters, prefer white or heavy-metal-free formulations. Finally, a bioceramic seal is only as durable as the coronal restoration above it: a sound, well-sealed restoration is essential to prevent leakage and protect the result.

Choosing a material — practical questions

- Is the site wet, and will it stay wet? Calcium-silicate materials tolerate moisture well.
- How fast must it set? Favour Biodentine or fast-set repair materials where time is limited.
- Is aesthetics at stake? Avoid bismuth- and iron-oxide-containing products near visible tissue.
- Is bulk strength or a thin sealing film required? Choose a cement or a sealer accordingly.

6.5 Limitations and precautions

For all their advantages, bioceramics have real constraints. Like all ceramics they are brittle, with limited crack resistance. Some formulations discolour tooth or soft tissue through bismuth or iron oxide. Set material can be difficult to remove if retreatment is later needed. Slow-setting products demand patience or protective measures, and cost can exceed that of traditional materials. Above all, the clinical evidence base, though growing, still lacks the long-term randomised trials that would place many indications beyond doubt. These limitations do not diminish the value of the materials, but they define where care and judgement are required.

CHAPTER SEVEN

Dental Adhesives

Bonding restorations to enamel, dentine, ceramics — and bioceramics

About this chapter

This chapter draws on the adhesion literature — in particular the account of enamel and dentine bonding by Breschi, Ferracane, Cadenaro, Mazzoni and Hilton in Summitt's Fundamentals of Operative Dentistry (4th ed.) and the primary studies cited there. Key sources are given as author–year citations and listed under 'Additional references' at the end of the book.

Almost every modern restoration is held in place, at least in part, by adhesion. Where earlier dentistry relied on mechanical retention — undercuts, pins, cements that merely filled a gap — adhesive dentistry bonds the restoration to the tooth, allowing more conservative preparations, better sealing and tooth-coloured materials. For the endodontically treated tooth in particular, a reliable adhesive seal over the access cavity is what protects the bioceramic root filling beneath it. This chapter explains how dental adhesives work and where they intersect with bioceramic materials.

7.1 Why adhesion matters

Adhesion serves three linked goals: retention of the restoration, a marginal seal against bacterial leakage, and reinforcement of the remaining tooth structure. A tooth restored with a well-bonded material behaves more like an intact tooth and is less prone to leakage and secondary caries. Because leakage is the common pathway to failure of both restorations and root-canal treatments, the quality of the adhesive seal is not a cosmetic detail but a determinant of long-term success.

7.2 The two substrates: enamel and dentine

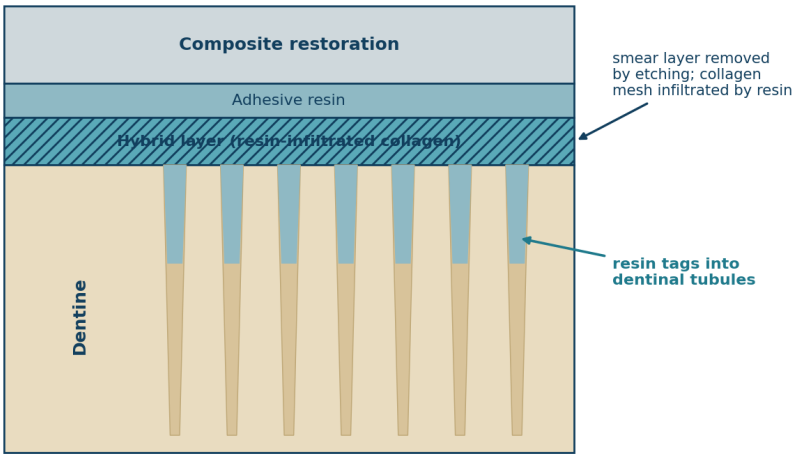
Enamel and dentine could hardly be more different, and bonding to each is a distinct problem. Enamel is highly mineralised, relatively dry and uniform; etching it with phosphoric acid — the technique first demonstrated by Buonocore in 1955 (Buonocore, 1955) — dissolves a few micrometres of mineral to create a microporous, high-energy surface into which fluid resin flows and, once cured, locks as countless microscopic tags. Enamel bonding is therefore strong, durable and predictable.

Dentine is the harder problem. It is less mineralised, intrinsically wet (threaded by fluid-filled tubules connected to the pulp), and rich in collagen. Cutting dentine also leaves a smear layer — a micrometre-thick film of cut debris that plugs the tubule mouths. A dentine adhesive must manage this moisture, deal with the smear layer, and bond to a surface that is part mineral and part organic. The solution is not glue but micromechanical infiltration, embodied in the hybrid layer.

7.3 The hybrid layer

When dentine is conditioned, the surface mineral is removed to expose a scaffold of collagen fibrils; a fluid resin then infiltrates this scaffold and is polymerised in place. The resulting hybrid layer — resin entangled within demineralised collagen, continuous with resin tags that extend into the tubules — is the true basis of dentine bonding. First described by Nakabayashi and co-workers in 1982 (Nakabayashi et al., 1982), it explains why technique matters so much: the resin must infiltrate to the full depth that the acid demineralised, or an unprotected, weak zone of bare collagen is left behind.

The adhesive interface: how resin bonds to dentine



Micromechanical interlock — not glue — is the basis of modern dentine bonding.

Figure 7.1 Dentine bonding is micromechanical: resin infiltrates a demineralised collagen mesh to form the hybrid layer and extends as tags into the tubules.

7.4 Etching strategies and adhesive families

Adhesive systems differ chiefly in how they treat the smear layer and deliver the resin. Two philosophies dominate. Etch-and-rinse (total-etch) systems apply phosphoric acid first, which is then rinsed away together with the smear layer, before priming and bonding; they come in three-step (etch, prime, bond) and two-step (etch, then combined prime-and-bond) forms. Self-etch systems instead use acidic monomers that condition and prime simultaneously and incorporate the smear layer rather than removing it; they come in two-step and one-step (all-in-one) forms.

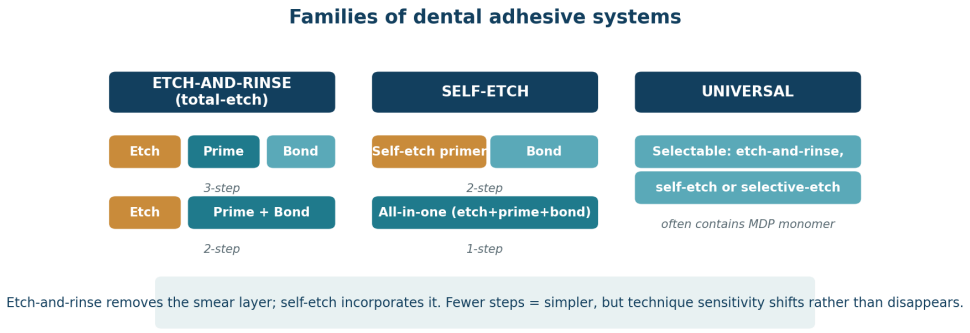


Figure 7.2 The main adhesive families. Fewer steps make a system simpler to use but shift, rather than remove, its technique sensitivity.

More recently, universal (multi-mode) adhesives allow the clinician to choose the etching mode — etch-and-rinse, self-etch, or selective enamel etching — with a single product. Many contain the functional monomer MDP (10-methacryloyloxydecyl dihydrogen phosphate), which forms a stable chemical bond to calcium in hydroxyapatite and also bonds to zirconia, widening the range of surfaces a single bottle can address. In general, etch-and-rinse gives the strongest enamel bond, while self-etch is less technique-sensitive on dentine and causes less post-operative sensitivity; selective enamel etching with a universal adhesive seeks the best of both.

7.5 Bonding to ceramics, zirconia and metal

Adhesion is not limited to tooth tissue. Each restorative substrate needs its own surface strategy. Glass-based ceramics (feldspathic, leucite, lithium disilicate) are etched with hydrofluoric acid to roughen the glassy phase, then treated with a silane coupling agent that chemically links the ceramic to the resin cement. Zirconia, having no glassy phase, does not respond to hydrofluoric acid; it is instead roughened by airborne-particle abrasion and primed with an MDP-containing primer that bonds to the zirconium oxide. Metal substrates rely on abrasion and dedicated metal primers.

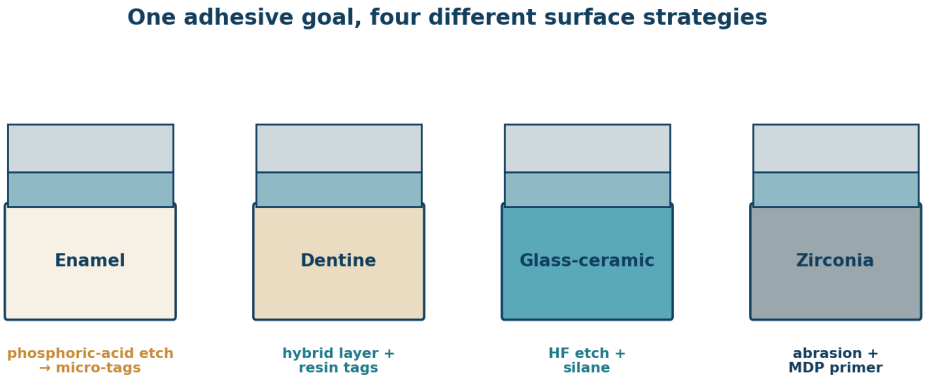


Figure 7.3 A single objective — a durable resin bond — met by four different surface treatments, from acid etching to MDP priming.

7.6 Durability: why bonds age

The immediate bond strength of modern adhesives is high; the challenge is keeping it. Over months and years the hybrid layer is attacked on two fronts. Hydrolysis — water slowly breaking down resin and any exposed collagen — degrades the interface, a process visible as nanoleakage (tracer penetration into nanometre-scale gaps; Sano et al., 1995). At the same time, the tooth's own enzymes, the matrix metalloproteinases and cysteine cathepsins, are activated by acidic conditioning and digest unprotected collagen within the hybrid layer (Pashley et al., 2004).

Several strategies slow this ageing: functional monomers such as MDP that form more hydrolytically stable chemical bonds; enzyme inhibitors (chlorhexidine is the classic example; Hebling et al., 2005; Carrilho et al., 2007) applied to the conditioned dentine; careful moisture control and

thorough light-curing; and, not least, a sound restoration that seals the margins and keeps the interface away from oral fluids. The recurring theme — familiar from bioceramics — is that a good seal protects everything beneath it.

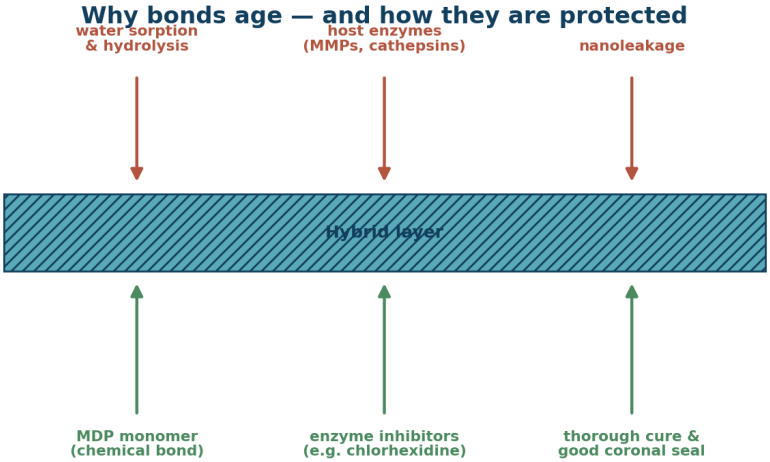


Figure 7.4 The adhesive bond ages through hydrolysis, enzyme activity and nanoleakage; stable monomers, enzyme inhibitors and a good seal all help to preserve it.

7.7 Adhesives and bioceramics

The two worlds of this book — adhesives and bioceramics — meet at several clinically important points. First, the coronal seal of an endodontically treated tooth is usually an adhesive restoration; however good the bioceramic root filling, its long-term success depends on a well-bonded restoration above it that keeps bacteria out. Second, clinicians increasingly place a calcium-silicate liner or dentine substitute (such as Biodentine) beneath a bonded composite; here the adhesive must bond reliably to the set bioceramic surface.

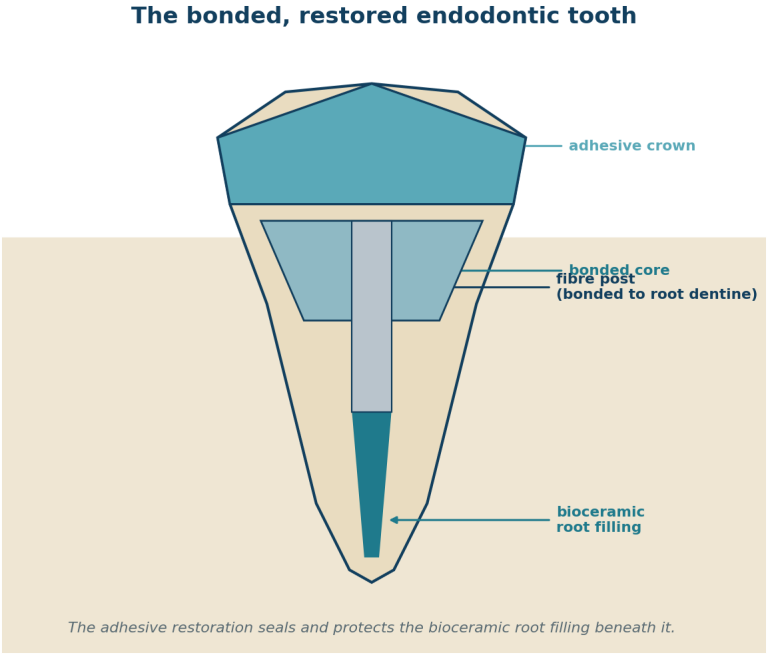


Figure 7.5 In the restored endodontic tooth the adhesive core, post and crown together form the coronal seal that protects the bioceramic root filling from bacterial leakage.

Clinical note — bonding over calcium-silicate materials

- Calcium-silicate materials set and mature over time; bonding a composite too early, to an immature surface, can give a weaker bond.
- Where the restorative plan allows, letting the material mature (or using a two-visit approach) tends to improve adhesion over Biodentine and MTA.

Third, chemical interactions matter. Eugenol-containing materials (such as traditional zinc-oxide-eugenol cements) can inhibit the polymerisation of resin and weaken an adhesive bond, which is one reason calcium-silicate and resin-compatible materials are often preferred when an adhesive restoration will follow. Finally, the field is moving toward bioactive, ion-releasing restorative materials and adhesives that aim to combine the sealing of adhesion with the calcium- and phosphate-releasing, remineralising behaviour characteristic of bioceramics — a convergence of the two material families.

7.8 Resin cements and luting indirect restorations

Indirect restorations — inlays, onlays, veneers, crowns and bridges — are attached to the tooth with a luting cement, and here adhesion is again central. Resin cements, used with an adhesive and the appropriate surface treatment of the restoration, give the strongest and best-sealing results and are essential for materials that rely on bonding for strength, such as thin veneers and glass-ceramics. They are broadly grouped as adhesive resin cements (used with a separate etch-and-rinse or self-etch adhesive) and self-adhesive resin cements, the latter simplifying the procedure by combining conditioning and luting in one material at some cost in bond strength. As with direct bonding, the choice balances bond strength, technique sensitivity and substrate: a bonded lithium-disilicate veneer demands a full adhesive protocol with silane, whereas a highly retentive zirconia crown may be luted more simply with an MDP-containing cement. Moisture control during luting is as decisive here as in direct restoration.

7.9 Bonding to root dentine: posts and cores

The endodontically treated tooth often lacks enough coronal structure to retain a restoration, and a post may be placed in the canal to retain a core that in turn supports a crown. Modern practice favours fibre posts bonded adhesively into the root, because their elasticity is closer to that of dentine than a rigid metal post and they distribute stress more favourably. Bonding into a root canal is, however, one of the most demanding adhesive situations: access is deep and narrow, moisture control is difficult, curing light barely reaches the apical portion, and residues of endodontic sealers — especially eugenol-containing ones — can impair the bond. This is a further reason resin-compatible bioceramic sealers are attractive when an adhesive post is anticipated. The unifying principle remains that the adhesive restoration and the bioceramic root filling protect the tooth only if the whole coronal seal holds.

7.10 Isolation, moisture control and common failures

More adhesive failures come from technique than from the choice of product. The single most important factor is isolation: a dry, saliva-free and blood-free field — most reliably a rubber dam — is fundamental, because contamination of an etched or primed surface ruins the bond. Beyond isolation, the common pitfalls are recognisable and avoidable:

1. Contamination of the bonding surface with saliva, blood or handpiece oil after conditioning.
2. Over-drying etched dentine (collapsing the collagen mesh), or leaving it visibly wet.
3. Incomplete evaporation of the adhesive solvent before curing.
4. Inadequate light-curing — too brief, too far away, or with a weak curing light.
5. Eugenol or other resin-inhibiting residues left on the bonding surface.
6. Bonding to an immature calcium-silicate surface before it has set and matured.

Because each of these is under the clinician's control, meticulous technique — not merely material selection — is what turns a good adhesive into a durable, well-sealed restoration.

7.11 Bioactive and self-adhesive restoratives: a convergence

The sharp line once drawn between 'adhesive' and 'bioactive' materials is blurring. Glass-ionomer and resin-modified glass-ionomer cements already combine adhesion to tooth structure with fluoride release and are often described as bioactive; newer ion-releasing composites, giomers and calcium- or phosphate-releasing restoratives extend the idea, aiming to seal a cavity adhesively while also buffering acid, releasing mineral-forming ions and encouraging remineralisation at the margin.

For the clinician this convergence is practical rather than abstract. A single material that bonds well and behaves bioactively could simplify the layered approach — bioceramic base, adhesive, composite — that is common today, and could improve the durability of the margin, the

commonest site of restoration failure. The evidence base for these newer materials is still maturing, and their bond strengths and long-term behaviour vary; but the direction of travel is clear, and it brings the two subjects of this book steadily closer together.

Chapter summary

- Adhesion provides retention, a marginal seal and reinforcement — and the seal protects the bioceramic work beneath it.
- Enamel bonds predictably after acid etching; dentine relies on the resin-infiltrated hybrid layer.
- Etch-and-rinse, self-etch and universal systems differ mainly in how they treat the smear layer.
- Ceramics need silane, zirconia needs MDP; bonds age by hydrolysis and enzyme activity.
- Over calcium-silicate materials, allow maturation for a better bond and avoid eugenol beforehand.

CHAPTER EIGHT

Reaction of Periodontal Tissues to Materials

How the periodontium responds — and why it favours bioceramics

About this chapter

This chapter draws together the tissue-response themes of the endodontic chapters. Its discussion of endodontic-sealer biocompatibility is supported by author–year citations to the primary literature reviewed by Dassieu (2023) and listed under ‘Additional references’; the broader histological trends are discussed, with their sources, in Chapters Two to Four.

Many of the procedures in this book place a material in direct, lasting contact with the periodontium — the ligament, cementum, bone and gingiva that surround and support the tooth. A root-end filling faces the apical periodontium; a perforation repair faces the lateral or furcal periodontium; a resorption repair or a subgingival margin faces the cervical periodontium and gingiva. How these tissues react to the material is therefore central to the outcome, and it is one of the strongest arguments for bioceramics.

8.1 The periodontium and where materials meet it

The periodontium has four components: the gingiva at the margin, the periodontal ligament (PDL) suspending the root, the cementum covering the root surface, and the alveolar bone forming the socket. Dental materials contact these tissues wherever the root surface or the pulp space is breached — at a resected apex, a perforation, a resorptive cavity, or a restoration margin that lies at or below the gingival crest.

The periodontium and where dental materials meet it

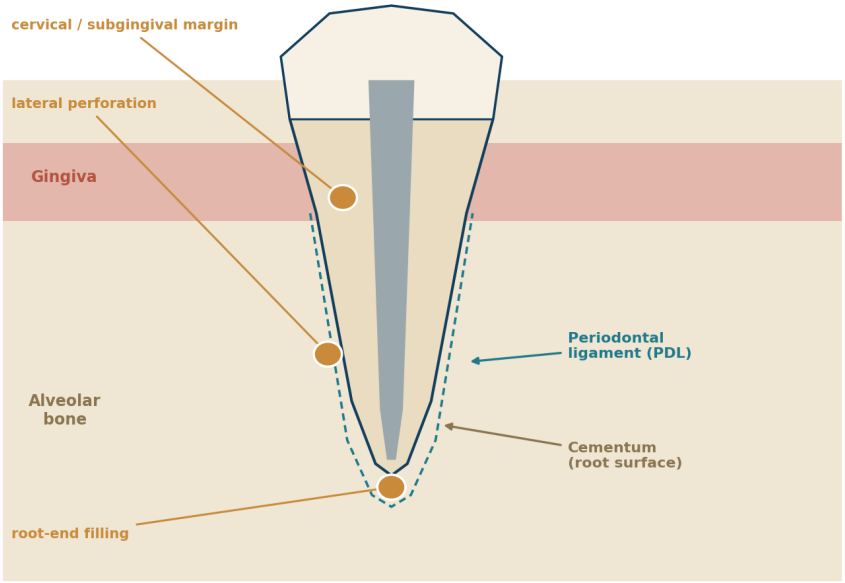


Figure 8.1 The supporting tissues of the tooth and the main points at which dental materials come into contact with them.

8.2 The spectrum of tissue response

A material in contact with periodontal tissue can provoke responses along a spectrum. At best it is biocompatible and even bioactive — tolerated without inflammation and actively encouraging repair, with new hard tissue forming on its surface. In the middle it is merely tolerated, separated from the tissue by a thin fibrous capsule. At worst it is irritant or cytotoxic, sustaining chronic inflammation, resorption and failure of healing. The clinical goal is to sit as far toward the bioactive end of this spectrum as possible, especially where the material will remain in contact with the tissue permanently.

8.3 How traditional materials behave

The materials that preceded bioceramics generally sit in the middle or unfavourable part of the spectrum. Amalgam, long used for root-end fillings, is tolerated but does not encourage regeneration and can cause discolouration and corrosion products in the tissue. Zinc-oxide-eugenol materials (including IRM) release eugenol, which is irritant and cytotoxic

in the concentrations found at a fresh interface. Super-EBA and reinforced cements improved handling and sealing but remain essentially non-regenerative. Glass-ionomer cements are better tolerated and release fluoride, but do not reliably induce new cementum or bone against the periodontium. Against this background the arrival of materials that actively favour periodontal healing was a significant advance.

8.4 The favourable response to bioceramics

Calcium-silicate bioceramics elicit a notably favourable periodontal response. Histological studies of materials such as MTA repeatedly show three encouraging features: the deposition of new cementum directly on the material surface (cementogenesis), the re-establishment of an organised periodontal ligament, and healing of the adjacent alveolar bone — often with a hard-tissue barrier forming across the interface and minimal inflammatory infiltrate. In effect the material is not merely tolerated but is incorporated into the repairing periodontium.

Bioceramics invite hard-tissue repair: cementum, ligament and bone

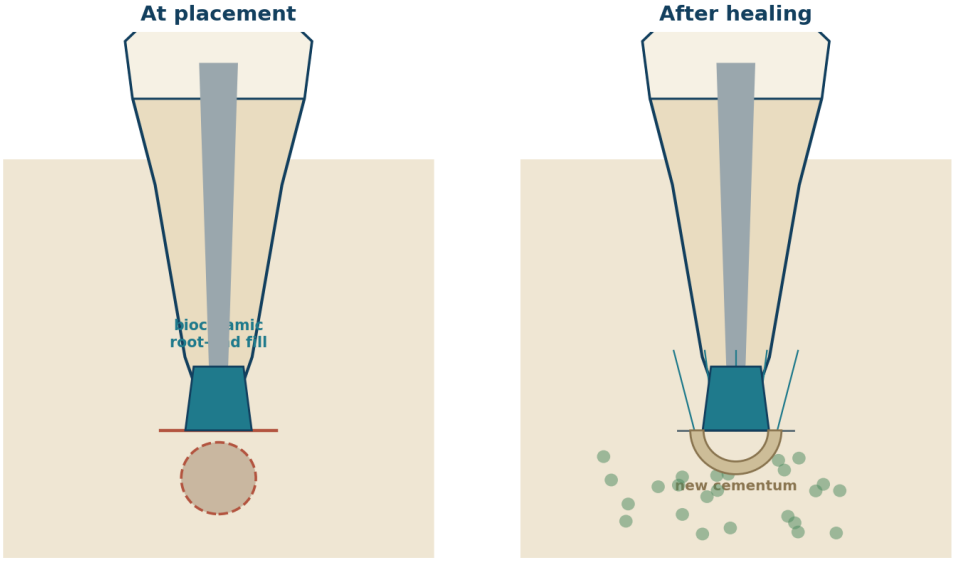


Figure 8.2 Over a bioceramic root-end filling, tissues tend to heal constructively: new cementum bridges the surface, ligament fibres reattach and bone fills in.

This behaviour is what allows bioceramics to change the prognosis of teeth that would once have been lost — perforated teeth, teeth needing apical surgery, teeth with resorptive defects communicating with the periodontium. The material does not simply plug a defect; it provides a surface on which the periodontium can rebuild.

8.5 Why: cellular and molecular mechanisms

The favourable response follows directly from the chemistry emphasised throughout this book. As the material sets and matures it maintains a high pH, releases calcium ions, and forms a surface apatite layer. This microenvironment supports the attachment and differentiation of the key periodontal cells: cementoblasts that lay down cementum, PDL fibroblasts that organise ligament fibres, and osteoblasts that form bone. The released calcium and the alkaline conditions up-regulate mineralisation signalling — including transforming growth factor-β and bone morphogenetic proteins — and markers of hard-tissue formation such as alkaline phosphatase and osteopontin.

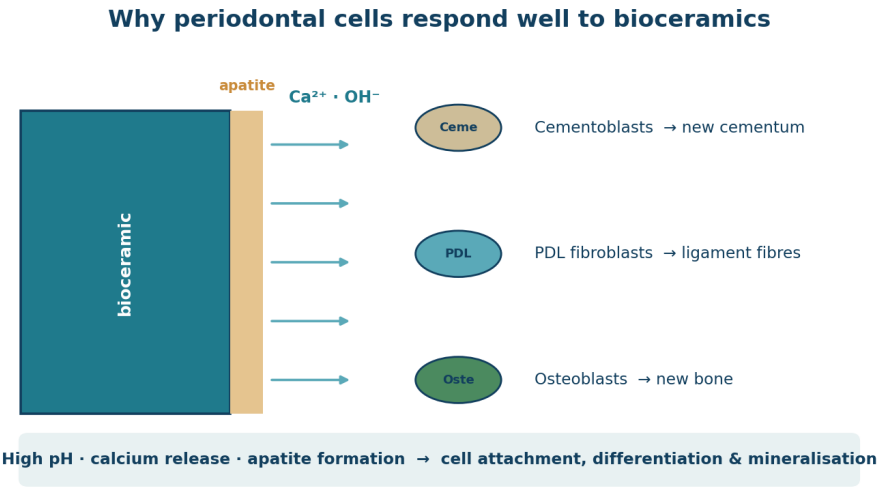


Figure 8.3 High pH, calcium release and apatite formation create a surface on which cementoblasts, ligament fibroblasts and osteoblasts attach, differentiate and mineralise.

8.6 Clinical settings

The same favourable interaction recurs across the clinical situations described in Chapter Four. In perforation repair, a bioceramic seal against the furcal or lateral periodontium limits inflammation and supports bone and cementum repair, where older materials invited attachment loss. In root-end filling, it promotes periradicular regeneration with new cementum and ligament over the resected surface. In invasive cervical resorption and other defects that communicate with the periodontium, its tolerance of moisture and its regenerative surface make it well suited to restoring the lost tissue. In regenerative endodontics, the coronal bioceramic plug sits against tissues from which new vital tissue is expected to grow.

8.7 A comparative view

Bringing these observations together, the materials used against the periodontium can be ranked qualitatively by the tissue response they tend to provoke. Calcium-silicate bioceramics and Biodentine sit at the favourable end, glass ionomer in the middle, and eugenol-based and corrosion-prone materials toward the less favourable end. The comparison below is a synthesis of histological trends rather than a quantitative scale, but the direction is consistent across studies.

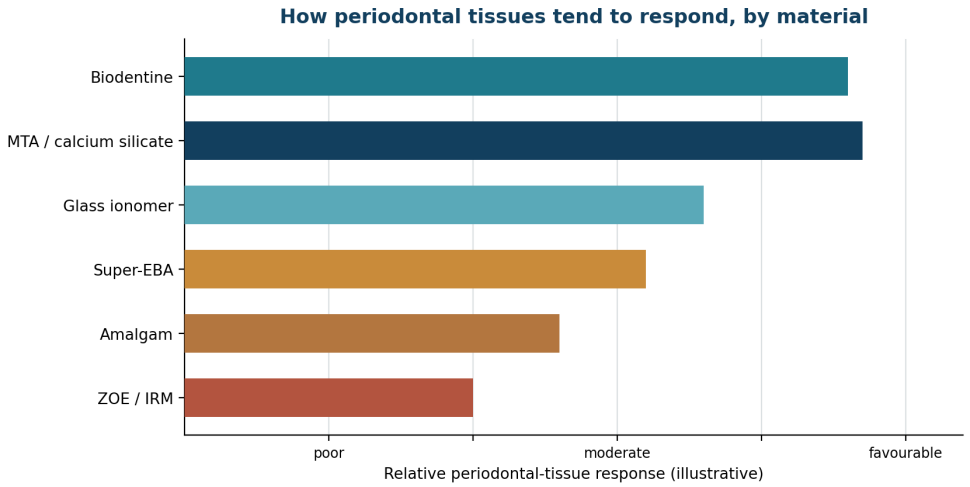


Figure 8.4 A qualitative synthesis: periodontal tissues tend to respond more constructively to calcium-silicate bioceramics than to older root-end and repair materials.

8.8 Soft-tissue and aesthetic considerations

Two practical caveats deserve mention. First, some bioceramics can discolour the tooth and the overlying gingiva through their bismuth- or iron-oxide radiopacifiers; near the visible gingival margin, white or heavy-metal-free formulations are preferred. Second, where a material or restoration margin lies close to the gingiva, respect for the biologic width — the soft-tissue attachment coronal to the bone — remains essential; even the most biocompatible material will not compensate for a margin that violates the attachment and provokes chronic inflammation.

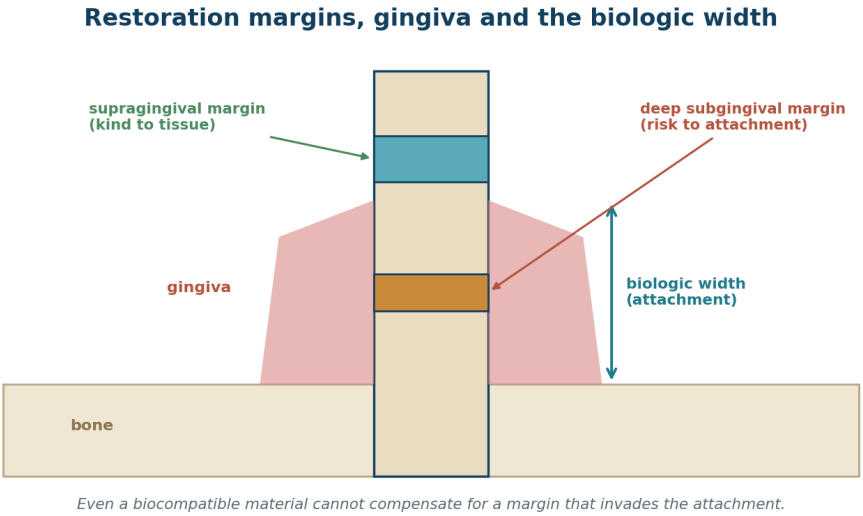


Figure 8.5 A supragingival margin is kind to the tissues; a deep subgingival margin that invades the biologic width provokes inflammation regardless of how biocompatible the material is.

8.9 The peri-implant interface

The reaction of tissues to materials is not confined to natural teeth. A dental implant sits in the same periodontal environment, and its success depends on two interfaces: the bone–implant interface, where osseointegration binds living bone directly to the surface (discussed in Chapter Five), and the soft-tissue interface, where the gingiva forms a seal around the abutment. Bioceramics contribute at both. Bioactive coatings such as hydroxyapatite encourage bone to grow onto the implant, while ceramic (zirconia) abutment surfaces are associated with a favourable soft-tissue response and low plaque affinity, helping to maintain the peri-implant seal. The same principle applies as around a tooth: a stable, well-adapted, biocompatible surface at the tissue margin protects the supporting structures, whereas roughness, contamination or a compromised seal invites inflammation — peri-implant mucositis and, if it progresses, peri-implantitis. Calcium-phosphate and bioactive-glass grafts further support the bone that must surround and stabilise the implant.

8.10 Endodontic sealers and the periodontal response

Nowhere is the difference between material families clearer than in the behaviour of root-canal sealers, which are extruded or diffuse to the apical and lateral periodontium and remain there in intimate contact with the tissues. The biocompatibility of a sealer is therefore a determinant of periradicular healing, and it has been studied by comparing the long-established epoxy-resin sealers (typified by AH Plus) with the newer calcium-silicate-based sealers on two fronts: the inflammatory response they provoke and the osteogenic (bone- and cementum-forming) potential they support (Dassieu, 2023).

On the first front, the literature consistently reports a lower inflammatory response to calcium-silicate sealers than to epoxy-resin sealers, whose freshly set surfaces tend to provoke a stronger initial reaction; cytotoxicity generally falls as calcium-silicate materials set and mature (Zhou et al., 2015; Saygili et al., 2017; Kapralos et al., 2022; Özdemir and Kopac, 2022). Subcutaneous-implantation and histological studies show the same direction of effect in vivo (Kohsar et al., 2022).

On the second front, calcium-silicate sealers show a positive osteogenic potential — up-regulating mineralisation markers and encouraging hard-tissue formation — where epoxy-resin sealers tend to be neutral or negative. Comparative laboratory studies on periodontal-ligament and bone-forming cells report greater expression of osteogenic markers and mineralisation with calcium-silicate sealers (Giacomino et al., 2019; Lee et al., 2019; Alves Silva et al., 2020), consistent with systematic and narrative reviews of the class (Donnermeyer et al., 2019; Lim et al., 2020). The mechanism is the one described earlier in this chapter: calcium release, alkalinity and apatite formation create a surface that periodontal cells tolerate and build upon.

What this means clinically

- Where a sealer may reach the periodontium, calcium-silicate materials tend to provoke less inflammation and better support hard-tissue healing than epoxy-resin sealers.
- The evidence is largely laboratory and animal-based; long-term clinical comparisons are still limited.

8.11 Evidence and gaps

The evidence that bioceramics favour periodontal healing is strong at the histological and animal level and is supported by clinical healing of periradicular and periodontal defects. As elsewhere in this field, the principal gap is long-term, randomised clinical data comparing materials on periodontal outcomes over many years. The consistency of the biological signal, however, gives clinicians good reason to prefer bioceramics wherever a material will meet the periodontium.

Chapter summary

- Many endodontic procedures place a material permanently against the periodontium; the tissue response drives the outcome.
- Traditional materials are tolerated at best; eugenol- and corrosion-based ones can be irritant.
- Bioceramics favour cementogenesis, ligament reattachment and bone healing with little inflammation.
- The mechanism is the familiar one: high pH, calcium release and apatite formation guiding periodontal cells.
- Prefer white / heavy-metal-free materials near the gingival margin, and respect the biologic width.

Conclusion

Introduced clinically in the 1990s as a root-end filling, mineral trioxide aggregate opened a field. Its uses have broadened continuously, and a growing family of bioceramic and hybrid materials has followed it into practice. These materials share a common calcium-silicate chemistry and a common signature — high alkalinity, calcium release and apatite formation — but differ in the mechanical properties that flow from their composition, above all compressive strength and setting time.

What unites them is biocompatibility and outstanding physicochemical behaviour, and a demonstrated capacity to overcome limitations of earlier endodontic materials. In doing so they have widened the dentist's reach: teeth with open apices, candidates for pulp capping and pulpotomy, retrograde fillings after apicoectomy, and the repair of perforations and resorptions can now be treated more predictably than before.

This expanded edition has set that endodontic story within a wider frame. Bioceramics belong to a family of dental materials that reaches from the all-ceramic crown to the bone graft, and their behaviour is best understood alongside the two disciplines that meet them most closely: adhesion, which provides the coronal seal that protects the bioceramic work beneath it and increasingly shares its bioactive ambitions; and the response of the periodontium, which is where the regenerative promise of bioceramics is most visibly realised, in new cementum, reattached ligament and healing bone. Seen together, these threads describe a shift in dentistry from materials that merely fill space to materials that seal, bond and help tissue to rebuild.

The evidence gap

- In-vitro and animal evidence for bioceramics is strong and consistent.
- Long-term, randomised, double-blind clinical trials remain scarce for most indications.
- Robust, well-designed studies with adequate follow-up are the priority for the field.

The promise of these in-vitro results now needs confirmation from long-term randomised, double-blind controlled trials over clinically meaningful periods. For most indications the high-quality clinical literature remains sparse. Closing that gap — with well-designed studies and adequate follow-up — is the task that will determine how far bioceramics reshape endodontic practice in the years ahead.

APPENDIX

Glossary of key terms

Term	Meaning
Apexification	Inducing a calcified barrier across an open apex, or continued apical development, in a necrotic immature tooth.
Bioactive material	A material that forms direct biochemical bonds with living tissue, typically via surface apatite formation.
Bioinert material	A material that coexists with tissue with minimal reaction, separated by a thin fibrous layer.
Bioresorbable material	A material gradually resorbed and replaced by host tissue during healing.
Calcium silicate hydrate (C–S–H)	The principal binding phase formed when calcium-silicate cements hydrate.
Ettringite	A calcium-aluminate-sulfate phase formed in the second (aluminate) hydration reaction of MTA.
Hydroxyapatite	The calcium-phosphate mineral of bone and tooth; its formation underlies bioceramic tissue induction.
Invasive cervical resorption (ECR)	External resorption starting below the epithelial attachment; graded I–IV (Heithersay).
Push-out bond strength	A laboratory measure of a material's resistance to displacement from the canal wall.
Radiopacity	Ability to attenuate X-rays, expressed as equivalent mm of aluminium (ISO 6876 minimum 3 mm).
Regenerative endodontics (RET)	Biologically based procedures aiming to replace the pulp–dentine complex and continue root development.
Retrograde (root-end) filling	A seal placed from the resected apex during endodontic microsurgery.
Tertiary (reparative) dentine	New dentine laid down by odontoblast-like cells in response to injury or a capping material.
Bioactive glass	A glass (e.g. 45S5) that forms a surface apatite layer bonding to tissue; used in grafts and remineralising agents.
CAD/CAM	Computer-aided design and manufacture; digital design and milling of ceramic restorations.
Dentine substitute	A material (e.g. Biodentine) used to replace missing dentine as a base or bulk restoration.

Term	Meaning
Glass-ceramic	A ceramic containing crystalline particles in a glassy matrix (e.g. leucite, lithium disilicate).
Guided bone regeneration (GBR)	Use of a barrier membrane to let slow-growing bone fill a defect undisturbed by soft tissue.
Osseointegration	Direct structural and functional connection between living bone and an implant surface.
Osteoconduction	Provision of a passive scaffold along which host bone grows.
Osteoinduction	Active recruitment and differentiation of progenitor cells into bone-forming cells.
Sintering	Heat treatment that fuses ceramic particles into a dense solid (e.g. for zirconia).
Transformation toughening	Crack-arresting mechanism in zirconia where grains change phase at a crack tip.
Y-TZP	Yttria-stabilised tetragonal zirconia polycrystal — the strong oxide ceramic used for crowns, bridges and implants.
Hybrid layer	Zone of resin-infiltrated demineralised dentine collagen that is the basis of dentine bonding.
Smear layer	Micrometre-thick film of cut debris left on dentine after preparation.
Etch-and-rinse	Adhesive approach that removes the smear layer with phosphoric acid before priming and bonding.
Self-etch	Adhesive approach using acidic monomers that condition and prime while incorporating the smear layer.
Universal adhesive	A multi-mode adhesive usable in etch-and-rinse, self-etch or selective-etch mode; often contains MDP.
MDP	10-methacryloyloxydecyl dihydrogen phosphate — a monomer that bonds chemically to hydroxyapatite and zirconia.
Silane	Coupling agent that chemically links a resin cement to an etched glass-ceramic surface.
Nanoleakage	Nanometre-scale tracer penetration marking degradation within the adhesive interface.

Term	Meaning
Resin cement	Adhesive luting material used to bond indirect restorations.
Periodontium	The supporting tissues of the tooth: gingiva, periodontal ligament, cementum and alveolar bone.
Cementogenesis	Formation of new cementum on a root or material surface.
Biologic width	The soft-tissue attachment (junctional epithelium and connective tissue) coronal to the alveolar bone.

APPENDIX

Abbreviations and acronyms

This list expands the abbreviations and acronyms used throughout the book.

Abbreviation	Term in full
MTA	Mineral trioxide aggregate
WMTA / GMTA	White / grey MTA
ProRoot	ProRoot MTA (Dentsply)
ERRM	EndoSequence Root Repair Material
BC	Bioceramic (e.g. BC Sealer)
iRoot SP / BP / FS	iRoot sealer / putty / fast-set repair
CEM	Calcium-enriched mixture cement
C–S–H	Calcium-silicate hydrate
$Ca(OH)_2$	Calcium hydroxide
HA	Hydroxyapatite
HCA	Hydroxy-carbonate apatite
TCP / β-TCP	Tricalcium phosphate / beta form
BCP	Biphasic calcium phosphate
DPC	Direct pulp capping
RET	Regenerative endodontic treatment
ICR / ECR	Internal / external (invasive) cervical resorption
GBR	Guided bone regeneration
Y-TZP	Yttria-stabilised tetragonal zirconia polycrystal
CAD/CAM	Computer-aided design / manufacture
GIC / RMGI	Glass-ionomer / resin-modified GIC
DPBS	Dulbecco's phosphate-buffered solution
ISO	International Organization for Standardization

Abbreviation	Term in full
AAE	American Association of Endodontists
TGF-β	Transforming growth factor beta
MTT / XTT	Colorimetric cell-viability assays
RUNX2	Runt-related transcription factor 2
MDP	10-methacryloyloxydecyl dihydrogen phosphate
HEMA	2-hydroxyethyl methacrylate
HF	Hydrofluoric acid
MMP	Matrix metalloproteinase
PDL	Periodontal ligament
ZOE	Zinc-oxide-eugenol
CEJ	Cemento-enamel junction

APPENDIX

Bioceramic materials at a glance

Material	Type	Main use(s)	Set time	Radiopacity (mm Al)	Notable feature
MTA (ProRoot)	Cement	Root-end fill, perforation, apexification, pulp cap	~165 min	~7.2	Gold standard; long set; can discolour
MTA Angelus	Cement	As MTA	~14 min	lower	Faster set; finer particle size
Biodentine	Cement	Dentine substitute, perforation, pulp cap	~10–12 min	>3	Fast set; good aesthetics; high Ca release
ERRM / iRoot BP	Cement (putty/paste)	Root repair, root-end fill	~4 h	high	Pre-mixed; handling ease
iRoot FS	Cement (fast-set)	Root repair	~1 h	high	Improved handling, faster set
BC Sealer / iRoot SP	Sealer	Canal obturation	water-dependent	3.83	Pre-mixed; high pH; hard to remove
MTA Fillapex	Hybrid sealer	Canal obturation	—	below AH Plus	High flow; higher solubility
MTA Plus	Cement / sealer	Root-end fill, sealing	tunable	similar MTA	Finer, lower-cost MTA; gel or liquid
Hydroxyapatite	Bioceramic	Bone graft, implant coating	—	high	Osteoconductive; slow resorption

Material	Type	Main use(s)	Set time	Radiopacity (mm Al)	Notable feature
β-TCP / biphasic	Bioceramic	Bone graft (GBR, sinus)	—	mod.	Resorbable; replaced by host bone
Bioactive glass	Bioceramic	Graft, remineralisation	—	low	Forms apatite; desensitising
Zirconia (Y-TZP)	Structural ceramic	Crowns, bridges, implants	(sintered)	—	Very strong; transformation toughening

Values are representative figures drawn from the studies cited in earlier chapters and from established materials data; they are for orientation, not specification.

APPENDIX

Material-selection quick guide

This guide maps common clinical situations to the bioceramic materials most often chosen for them, with a key caution for each. It summarises the reasoning developed in Chapters Two to Four and is intended as an aide-mémoire, not a protocol; material choice must always follow individual case assessment and current guidelines.

Clinical situation	Commonly chosen material(s)	Key caution
Root-end filling (apicoectomy)	MTA; fast-set repair putty (ERRM / iRoot FS)	Long MTA set in a moist surgical field — a fast-set material may help
Furcal / lateral perforation	MTA; Biodentine	Avoid bismuth/iron-oxide products near the gingival margin (staining)
Perforating internal resorption	Bioceramic cement (MTA / Biodentine)	Only needed once the root is perforated; seal to sound dentine
Invasive cervical resorption (Class III–IV)	Bioceramic cement; RRM/FS putty	Biodentine is less radiopaque — harder to assess on radiographs
Direct pulp cap / partial pulpotomy	MTA; Biodentine	Ensure haemostasis and a bacteria-tight coronal seal
Apical plug in an open apex	MTA; MTA Plus; BC RRM-FS putty	Reinforce the thin root with a bonded coronal restoration
Regenerative endodontics (coronal plug)	White MTA	Use white MTA to limit crown discolouration
Canal obturation (sealer)	BC Sealer / iRoot SP	Dry the canal first; the set is water-dependent and hard to remove
Deep liner / dentine substitute	Biodentine; light-cured calcium-silicate liner	Not a substitute for a definitive restoration
Alveolar bone graft / ridge preservation	Hydroxyapatite; β-TCP; biphasic; bioactive glass	Match resorption rate to the site; consider a barrier membrane (GBR)

APPENDIX

Adhesive systems at a glance

A summary of the adhesive families discussed in Chapter Seven, with their steps and typical use. It is an orientation aid, not a substitute for the manufacturer's instructions, which must always be followed for a given product.

System	Steps	Smear layer	Notable points	Typical use
Etch-and-rinse, 3-step	Etch · prime · bond	Removed	Strong, well-studied enamel & dentine bond	Gold-standard bonding; demanding cases
Etch-and-rinse, 2-step	Etch · prime-and-bond	Removed	Simpler; slightly more technique-sensitive	Routine direct restorations
Self-etch, 2-step	Self-etch primer · bond	Incorporated	Less post-op sensitivity; milder etch of enamel	Dentine-bonded restorations
Self-etch, 1-step	All-in-one	Incorporated	Simplest; generally lowest, less durable bond	Quick or paediatric cases
Universal (multi-mode)	Selectable mode	Either	Contains MDP; bonds enamel, dentine, zirconia	Versatile everyday use; selective-etch enamel
Self-adhesive resin cement	One step (lute)	Incorporated	Convenient luting; lower bond than adhesive cement	Retentive crowns; posts

Remember

- Enamel bonds best after phosphoric-acid etching; glass-ceramics need silane; zirconia needs MDP.
- Whatever the system, isolation and moisture control decide the result.

APPENDIX

Periodontal response of common materials

A qualitative synthesis of how periodontal tissues tend to respond to materials placed against them (Chapter Eight). Trends are drawn from histological and animal studies; they indicate direction, not a quantitative ranking.

Material	Tissue response	Regeneration	Key caution
MTA (calcium silicate)	Favourable, low inflammation	Cementum, PDL & bone form on surface	Long set; grey forms discolour
Biodentine	Favourable	Supports hard-tissue repair	Less radiopaque than MTA
Bioceramic sealers / putties	Favourable	Biomineralisation at interface	Confirm set; hard to remove
Glass ionomer	Moderate, tolerated	Little periodontal regeneration	Not for deep periradicular repair
Super-EBA	Moderate	Non-regenerative	Handling-dependent seal
Amalgam	Tolerated	None; can discolour tissue	Corrosion products; being superseded
ZOE / IRM	Less favourable	Irritant while fresh	Eugenol cytotoxicity

Remember

- Prefer calcium-silicate bioceramics wherever a material will remain against the periodontium.
- Near the gingival margin choose white / heavy-metal-free formulations and respect the biologic width.

APPENDIX

Clinical protocols at a glance

The following outlines distil the procedures discussed in Chapter Four into concise sequences. They are teaching summaries, not a substitute for full clinical training, current guidelines or individual case assessment.

Root-end (retrograde) filling

1. Reflect a full-thickness flap and gain access to the periapical lesion under magnification.
2. Resect the apical 3 mm of root and curette the lesion.
3. Prepare a retrograde cavity along the canal axis with ultrasonic tips.
4. Dry the cavity; place and condense a bioceramic cement (e.g. MTA) as the root-end fill.
5. Confirm the seal, irrigate, reposition the flap and suture; review radiographically.

Note

MTA gives excellent sealing and periradicular regeneration; fast-set repair materials ease handling in a moist field.

Furcal / lateral perforation repair

1. Achieve haemostasis and isolate the perforation; control moisture.
2. If needed, place an internal matrix (e.g. resorbable collagen) to prevent extrusion.
3. Place a bioceramic cement (MTA or Biodentine) against the defect with light condensation.
4. Verify the seal; where slow-setting cement is used, protect with a moist pellet until set.
5. Restore the remaining access and provide a definitive coronal seal.

Note

Prefer white or heavy-metal-free materials near the gingival margin to avoid discolouration.

Apical plug (apexification) in an open-apex tooth

1. Disinfect the canal by gentle chemo-mechanical debridement; inter-visit calcium hydroxide if indicated.
2. Dry the canal and place an apical plug of bioceramic cement (~4 mm), condensed to the apex.
3. Confirm plug position and adaptation radiographically.
4. Backfill the remaining canal and place a bonded coronal restoration to reinforce the root.

Note

MTA and comparable cements give earlier filling and better fracture resistance than long-term calcium hydroxide.

Regenerative endodontic procedure (immature necrotic tooth)

1. Disinfect with copious irrigation and an intracanal medicament (e.g. triple antibiotic paste).
2. At the second visit, induce apical bleeding to form a blood clot, or place platelet-rich plasma as a scaffold.
3. Place a bioceramic cement (white MTA) as a coronal plug over the scaffold.
4. Seal with a definitive restoration; monitor for continued root development.

Note

Aims to regenerate — not merely replace — the pulp-dentine complex; outcomes require long-term follow-up.

Direct pulp capping / partial pulpotomy

1. Confirm a favourable diagnosis and achieve haemostasis at the exposure.
2. For partial pulpotomy, remove superficial unhealthy pulp to sound tissue.
3. Place a calcium-silicate material (MTA or Biodentine) directly over the pulp wound.
4. Seal with a well-adapted restoration to prevent bacterial leakage; review vitality.

Note

Calcium-silicate materials outperform calcium hydroxide for dentine-bridge quality and clinical success.

Adhesive restoration over a calcium-silicate base

1. Place and allow the calcium-silicate material (e.g. Biodentine) to set and begin to mature.
2. Isolate the tooth (rubber dam) and clean the surface, avoiding eugenol-containing materials beforehand.
3. Apply the chosen adhesive (a universal or etch-and-rinse system) per the manufacturer's instructions.
4. Build up with composite in increments, curing each thoroughly.
5. Check the margin and occlusion; a sound seal protects the bioceramic beneath.

Note

Where possible, delay definitive bonding until the calcium-silicate surface has matured to improve adhesion.

Bonded fibre post in an endodontically treated tooth

1. Confirm an adequate apical seal and remove gutta-percha/sealer to the planned post depth, leaving 4–5 mm apically.
2. Prefer a resin-compatible (non-eugenol) sealer where a bonded post is anticipated.
3. Isolate, condition the root canal and apply the adhesive; control moisture carefully in the deep canal.
4. Cement the fibre post with a (often dual-cure) resin cement and build the core.
5. Restore with a crown that provides a ferrule and a sound coronal seal.

Note

Bonding into the canal is technique-sensitive; isolation and eugenol-free residues are decisive.

Perforation repair protecting the periodontium

1. Locate the perforation and control haemorrhage and moisture.
2. If the defect is large or faces the periodontium, place an internal matrix to prevent extrusion.
3. Seal with a calcium-silicate bioceramic (MTA or Biodentine) adapted to sound margins.
4. Prefer white / heavy-metal-free material near the gingival margin to avoid discolouration.
5. Confirm the seal, then restore; respect the biologic width at the tissue margin.

Note

A biocompatible seal favours cementum and bone repair; it cannot compensate for a margin that invades the periodontal attachment.

References

The following bibliography is reproduced from the original text by the authors (Prince Abdulrahman Advanced Dental Institute, Riyadh, 2025) and lists the sources for the author–year citations used in Chapters One to Four. The background Chapters Five and Six do not draw on these references.

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About this edition

Bioceramic in Dentistry — Expanded & Illustrated Edition

This edition preserves the text, scholarship and full bibliography of the original work by Dr. Saad Alsaif and Dr. Khalid Al Otaibi (Prince Abdulrahman Advanced Dental Institute, Riyadh, 2026). The endodontic chapters and their citations are the authors' own work.

Added for this edition are twenty-one original schematic illustrations and comparative charts, section summaries and clinical call-outs, two background chapters placing endodontic bioceramics in the wider context of dentistry and materials science, and a set of quick-reference appendices. The illustrations are diagrammatic, not clinical images; the chart values are representative figures drawn from the cited literature; and the background chapters are written as established materials science rather than as new, separately-cited claims.

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